Artificial Intelligence

How do we define Artificial Intelligence?

ARTIFICIAL + INTELLIGENCE

refers to something that is made or **created by humans** rather than **occurring naturally**

refers to the ability to **learn**, **understand**, **reason**, and **solve problems**.

INFORMAL AND EPHEMERAL DEFINITION

"AI is the study of how to make computers do things, which, at the moment, people do better."

Formal and Official Definition

- Artificial Intelligence refers to the **simulation of human intelligence in machines** that are programmed to think, learn, and perform tasks **typically requiring human intelligence**.

- AI enables machines to perform tasks such as **problem-solving**, and **decision-making.**

History of AI

Milestones in AI Development

The Turing Test	ELIZA	The Deep Blue	IBM Watson	Rise of Generative AI
1950	**1966**	**1997**	**2011**	**2020**
Alan Turing introduces the Turing Test to determine if a machine can exhibit human-like intelligence.	An early chatbot developed to mimic conversation by recognizing keywords	defeats world chess champion Garry Kasparov.	wins a quiz show while outperforming human champions.	AI advancements in self-driving technology, healthcare, and NLP with tools like **GPT-4o, Llama**.

Some of the Task Domains of Artificial Intelligence

Mundane Tasks

- Perception
  - Vision
  - Speech
- Natural language
  - Understanding
  - Generation
  - Translation
- Commonsense reasoning
- Robot control

Formal Tasks

- Games
  - Chess
  - Backgammon
  - Checkers -Go
- Mathematics
  - Geometry
  - Logic
  - Integral calculus
  - Proving properties of programs

Expert Tasks

- Engineering
  - Design
  - Fault finding
  - Manufacturing planning
- Scientific analysis
- Medical diagnosis
- Financial analysis

What is an AI Technique?

One of the few hard and fast results to come out of the first three decades of AI research is that *intelligence requires knowledge*. To compensate for its one overpowering asset, indispensability, knowledge possesses some less desirable properties, including:

- It is voluminous.
- It is hard to characterize accurately.
- It is constantly changing.
- It differs from data by being organized in a way that corresponds to the ways it will be used.

So where does this leave us in our attempt to define AI techniques? We are forced to conclude that an AI technique is a method that exploits knowledge that should be represented in such a way that:

- The knowledge captures generalizations. In other words, it is not necessary to represent separately each individual situation. Instead, situations that share important properties are grouped together. If knowledge does not have this property, inordinate amounts of memory and updating will be required. So we usually call something without this property "data" rather than knowledge.
- It can be understood by people who must provide it. Although for many programs, the bulk of the data can be acquired automatically (for example, by taking readings from a variety of instruments), in many AI domains, most of the knowledge a program has must ultimately be provided by people in terms they understand.

1.3.1 Tic-Tac-Toe

In this section, we present a series of three programs to play tic-tac-toe. The programs in this series increase in:

- Their complexity
- Their use of generalizations
- The clarity of their knowledge
- The extensibility of their approach. Thus, they move toward being representations of what we call AI techniques.

Data Structures

Board A nine-element vector representing the board, where the elements of the vector correspond to the board positions as follows:

1	2	3
4	5	6
7	8	9

An element contains the value 0 if the corresponding square is blank, 1 if it is filled with an X, or 2 if it is filled with an O.

Movetable A large vector of 19,683 elements (3^9), each element of which is a nine-element vector. The contents of this vector are chosen specifically to allow the algorithm to work.

The Algorithm

To make a move, do the following:

1. View the vector Board as a ternary (base three) number. Convert it to a decimal number.
2. Use the number computed in step 1 as an index into Movetable and access the vector stored there.
3. The vector selected in step 2 represents the way the board will look after the move that should be made. So set Board equal to that vector.

Defining the problem as a State Space?

For the problem "Play chess," it is fairly easy to provide a formal and complete problem description. The starting position can be described as an 8×8 array where each position contains a symbol standing for the appropriate piece in the official chess opening position. We can define as our goal any board position in which the opponent does not have a legal move and his or her king is under attack. The legal moves provide the way of getting from the initial state to a goal state. They can be described easily as a set of rules consisting of two

describes the change to be made to the board position to reflect the move. There are several ways in which these rules can be written. For example, we could write a rule such as that shown in Fig. 2.1.

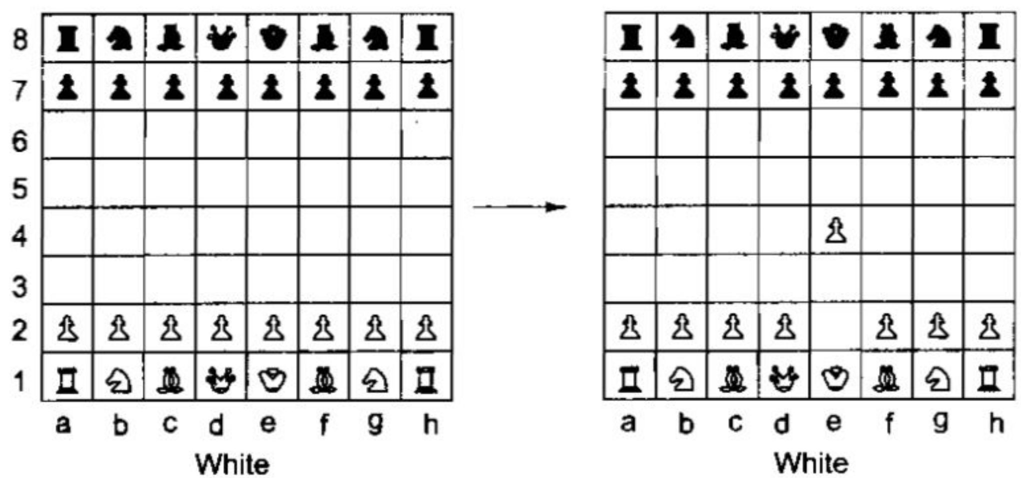


Fig. 2.1 *One Legal Chess Move*

serious practical difficulties:

- No person could ever supply a complete set of such rules. It would take too long and could certainly not be done without mistakes.
- No program could easily handle all those rules. Although a hashing scheme could be used to find the relevant rules for each move fairly quickly, just storing that many rules poses serious difficulties.

as shown in Fig. 2.2.[1] In general, the more succinctly we can describe the rules we need, the less work we will have to do to provide them and the more efficient the program that uses them can be.

White pawn at Square(file e, rank 2) AND Square(file e, rank 3) is empty AND Square(file e, rank 4) is empty	→	move pawn from Square(file e, rank 2) to Square(file e, rank 4)

Fig. 2.2 *Another Way to Describe Chess Moves*

We have just defined the problem of playing chess as a problem of moving around in a *slate space*, matrix to describe an individual state. The state space representation forms the basis of most of the AI methods we discuss here. Its structure corresponds to the structure of problem solving in two important ways:

- It allows for a formal definition of a problem as the need to convert some given situation into some desired situation using a set of permissible operations.
- It permits us to define the process of solving a particular problem as a combination of known techniques (each represented as a rule defining a single step in the space) and search, the general technique of exploring the space to try to find some path from the current state to a goal state. Search is a very important process in the solution of hard problems for which no more direct techniques are available.

In order to show the generality of the state space representation, we use it to describe a problem very different from that of chess.

A Water Jug Problem: You are given two jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug?

The state space for this problem can be described as the set of ordered pairs of integers (x, y), such that x = 0, 1, 2, 3, or 4 and y = 0, 1, 2, or 3; x represents the number of gallons of water in the 4-gallon jug, and y represents the quantity of water in the 3-gallon jug. The start state is (0, 0). The goal state is $(2, n)$ for any value of n (since the problem does not specify how many gallons need to be in the 3-gallon jug).

For the water jug problem, as with many others, there are several sequences of operators that solve the problem. One such sequence is shown in Fig. 2.4. Often, a problem contains the explicit or implied statement that the shortest (or cheapest) such sequence be found. If present, this requirement will have a significant effect on the choice of an appropriate mechanism to guide the search for a solution. We discuss this issue in Section 2.3.4.

Solution to Water Jug Problem

1	(x, y) if $x < 4$	$\rightarrow (4, y)$	Fill the 4-gallon jug
2	(x, y) if $y < 3$	$\rightarrow (x, 3)$	Fill the 3-gallon jug
3	(x, y) if $x > 0$	$\rightarrow (x - d, y)$	Pour some water out of the 4-gallon jug
4	(x, y) if $y > 0$	$\rightarrow (x, y - d)$	Pour some water out of the 3-gallon jug
5	$(x, y\}$ if $x > 0$	$\rightarrow (0, y)$	Empty the 4-gallon jug on the ground
6	(x, y) if $y > 0$	$\rightarrow (x, 0)$	Empty the 3-gallon jug on the ground
7	(x, y) if $x + y \geq 4$ and $y > 0$	$\rightarrow (4, y - (4 - x))$	Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full
8	(x, y) if $x + y \geq 3$ and $x > 0$	$\rightarrow (x - (3 - y), 3)$	Pour water from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full
9	(x, y) if $x + y \leq 4$ and $y > 0$	$\rightarrow (x + y, 0)$	Pour all the water from the 3-gallon jug into the 4-gallon jug
10	(x, y) if $x + y \leq 3$ and $x > 0$	$\rightarrow (0, x + y)$	Pour all the water from the 4-gallon jug into the 3-gallon jug
11	$(0, 2)$	$\rightarrow (2, 0)$	Pour the 2 gallons from the 3-gallon jug into the 4-gallon jug
12	$(2, y)$	$\rightarrow (0, y)$	Empty the 2 gallons in the 4-gallon jug on the ground

Gallons in the 4-Gallon Jug	Gallons in the 3-Gallon Jug	Rule Applied
0	0	
		2
0	3	
		9
3	0	
		2
3	3	
		7
4	2	
		5 or 12
0	2	
		9 or 11
2	0	

Fig. 2.4 *One Solution to the Water Jug Problem*

Problem Characteristics

most appropriate method (or combination of methods) for a particular problem, it is necessary to analyze the problem along several key dimensions:

- Is the problem decomposable into a set of (nearly) independent smaller or easier subproblems?
- Can solution steps be ignored or at least undone if they prove unwise?
- Is the problem's universe predictable?
- Is a good solution to the problem obvious without comparison to all other possible solutions?
- Is the desired solution a state of the world or a path to a state?
- Is a large amount of knowledge absolutely required to solve the problem, or is knowledge important only to constrain the search?
- Can a computer that is simply given the problem return the solution, or will the solution of the problem require interaction between the computer and a person?

What is an Expert System?

Jackson (1999) provides us with the following definition:

> An ***expert system*** is a computer program that represents and reasons with knowledge of some specialist subject with a view to solving problems or giving advice.

To solve expert-level problems, expert systems will need efficient access to a substantial domain ***knowledge base***, and a ***reasoning mechanism*** to apply the knowledge to the problems they are given. Usually they will also need to be able to explain, to the users who rely on them, how they have reached their decisions.

They will generally build upon the ideas of knowledge representation, production rules, search, and so on, that we have already covered.

Tasks of an Expert System

There are no fundamental limits on what problem domains an expert system can be built to deal with. Some typical existing expert system tasks include:

1. The interpretation of data

   Such as sonar data or geophysical measurements
2. Diagnosis of malfunctions

   Such as equipment faults or human diseases
3. Structural analysis or configuration of complex objects

   Such as chemical compounds or computer systems
4. Planning sequences of actions

   Such as might be performed by robots
5. Predicting the future

   Such as weather, share prices, exchange rates

Characteristics of an Expert System

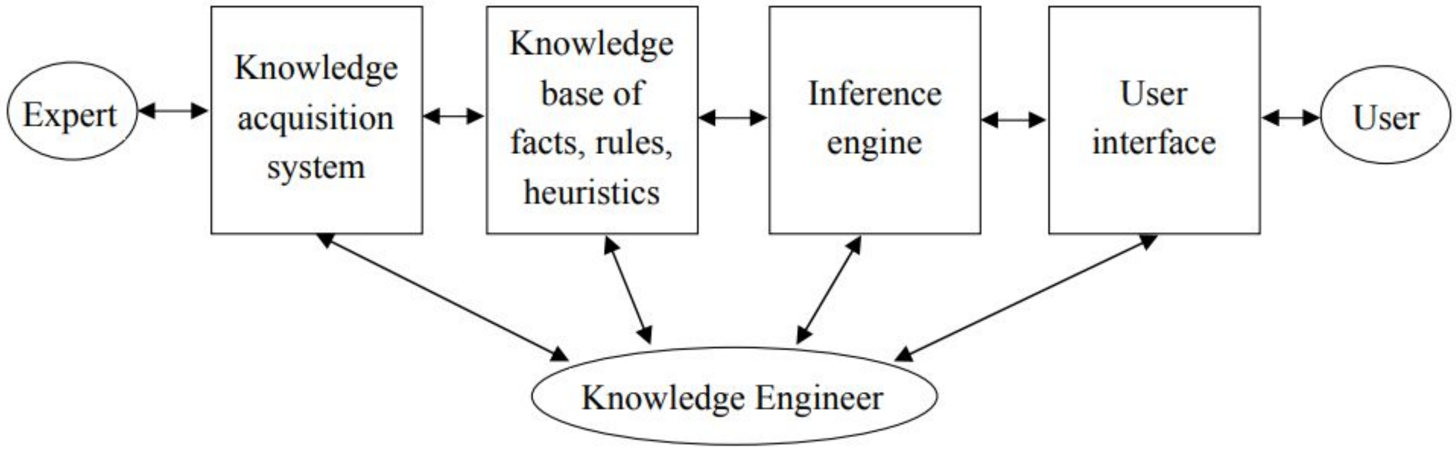
Expert systems can be distinguished from conventional computer systems in that:

1. They ***simulate human reasoning*** about the problem domain, rather than simulating the domain itself.
2. They perform ***reasoning*** over ***representations of human knowledge***, in addition to doing numerical calculations or data retrieval. They have corresponding distinct modules referred to as the ***inference engine*** and the ***knowledge base***.
3. Problems tend to be solved using ***heuristics*** (rules of thumb) or ***approximate methods*** or ***probabilistic methods*** which, unlike algorithmic solutions, are not guaranteed to result in a correct or optimal solution.
4. They usually have to provide ***explanations*** and ***justifications*** of their solutions or recommendations in order to convince the user that their reasoning is correct.

Note that the term Intelligent Knowledge Based System (IKBS) is sometimes used as a synonym for Expert System.

Architecture of Expert Systems

The process of building expert systems is often called ***knowledge engineering***. The ***knowledge engineer*** is involved with all components of an expert system:



Building expert systems is generally an iterative process. The components and their interaction will be refined over the course of numerous meetings of the knowledge engineer with the experts and users. We shall look in turn at the various components.

Knowledge Acquisition

The knowledge acquisition component allows the expert to enter their knowledge or expertise into the expert system, and to refine it later as and when required.

Historically, the knowledge engineer played a major role in this process, but automated systems that allow the expert to interact directly with the system are becoming increasingly common.

The knowledge acquisition process is usually comprised of three principal stages:

1. ***Knowledge elicitation*** is the interaction between the expert and the knowledge engineer/program to elicit the expert knowledge in some systematic way.
2. The knowledge thus obtained is usually stored in some form of human friendly ***intermediate representation***.
3. The intermediate representation of the knowledge is then compiled into an ***executable form*** (e.g. production rules) that the inference engine can process.

In practice, many iterations through these three stages are usually required!

Knowledge Elicitation

The knowledge elicitation process itself usually consists of several stages:

1. Find as much as possible about the problem and domain from books, manuals, etc. In particular, become familiar with any specialist terminology and jargon.

2. Try to characterise the types of reasoning and problem solving tasks that the system will be required to perform.

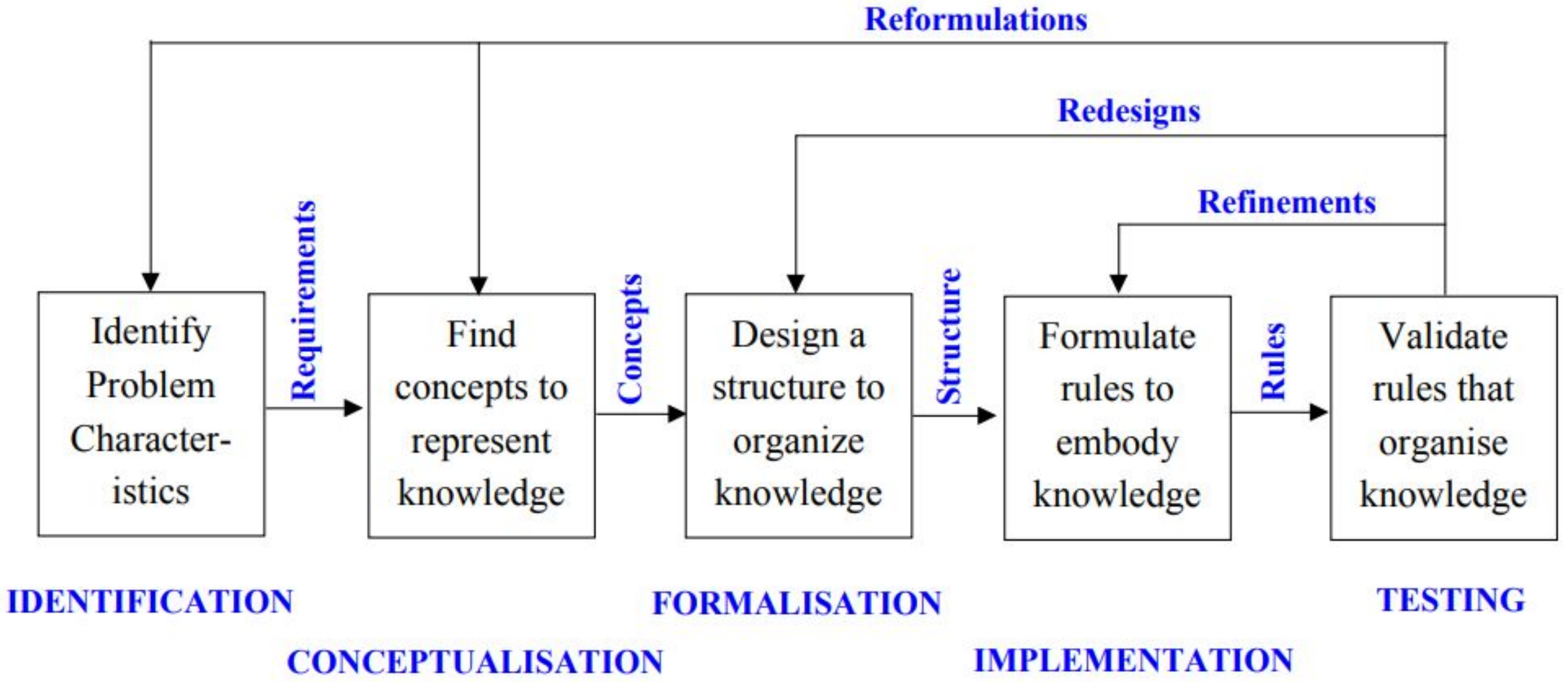
3. Find an expert (or set of experts) that is willing to collaborate on the project. Sometimes experts are frightened of being replaced by a computer system!

4. Interview the expert (usually many times during the course of building the system). Find out how they solve the problems your system will be expected to solve. Have them check and refine your intermediate knowledge representation.

This is a time intensive process, and ***automated knowledge elicitation*** and ***machine learning*** techniques are increasingly common modern alternatives.

Stages of Knowledge Acquisition

The iterative nature of the knowledge acquisition process can be represented in the following diagram (from Jackson, Section 10.1):



Representing the Knowledge

We have already looked at various types of knowledge representation. In general, the knowledge acquired from our expert will be formulated in two ways:

1. **Intermediate representation** – a structured knowledge representation that the knowledge engineer and expert can both work with efficiently.
2. **Production system** – a formulation that the expert system's inference engine can process efficiently.

It is important to distinguish between:

1. **Domain knowledge** – the expert's knowledge which might be expressed in the form of rules, general/default values, and so on.
2. **Case knowledge** – specific facts/knowledge about particular cases, including any derived knowledge about the particular cases.

The system will have the domain knowledge built in, and will have to integrate this with the different case knowledge that will become available each time the system is used.

Rules as a Knowledge Representation Technique

- The term rule in AI, which is the most commonly used type of knowledge representation, can be defined as an IF-THEN structure that relates given information or facts in the IF part to some action in the THEN part.
- A rule provides some description of how to solve a problem.
- Rule are relatively easy to create and understand
- Any rules consists of two parts: the IF part, called the *antecedent* (premise or condition) and the THEN part called the *consequent* (conclusion or action)

IF	<antecedent>
THEN	<consequent>

- A rule can have multiple antecedents joined by the keywords **AND (conjunction)**, **OR (disjunction)** or a combination of both.

IF	<antecedent 1>	IF	<antecedent 1>
AND	<antecedent 2>	OR	<antecedent 2>
:		:	
AND	<antecedent *n*>	OR	<antecedent *n*>
THEN	<consequent>	THEN	<consequent>

IF	the ‘traffic light’ is ‘green’
THEN	the action is go

IF	the ‘traffic light’ is ‘red’
THEN	the action is stop

The Inference Engine

We have already looked at production systems, and how they can be used to generate new information and solve problems.

Recall the steps in the basic Recognize Act Cycle:

1. ***Match*** the premise patterns of the rules against elements in the working memory. Generally the rules will be domain knowledge built into the system, and the working memory will contain the case based facts entered into the system, plus any new facts that have been derived from them.
2. If there is more than one rule that can be applied, use a ***conflict resolution*** strategy to choose one to apply. Stop if no further rules are applicable.
3. ***Activate*** the chosen rule, which generally means adding/deleting an item to/from working memory. Stop if a terminating condition is reached, or return to step 1.

Additional Problems

The Missionaries and Cannibals Problem

Three missionaries and three cannibals find themselves on one side of a river. They have agreed that they would all like to get to the other side. But the missionaries are not sure what else the cannibals have agreed to. So the missionaries want to manage the trip across the river in such a way that the number of missionaries on either side of the river is never less than the number of cannibals who are on the same side. The only boat available holds only two people at a time. How can everyone get across the river without the missionaries risking being eaten?

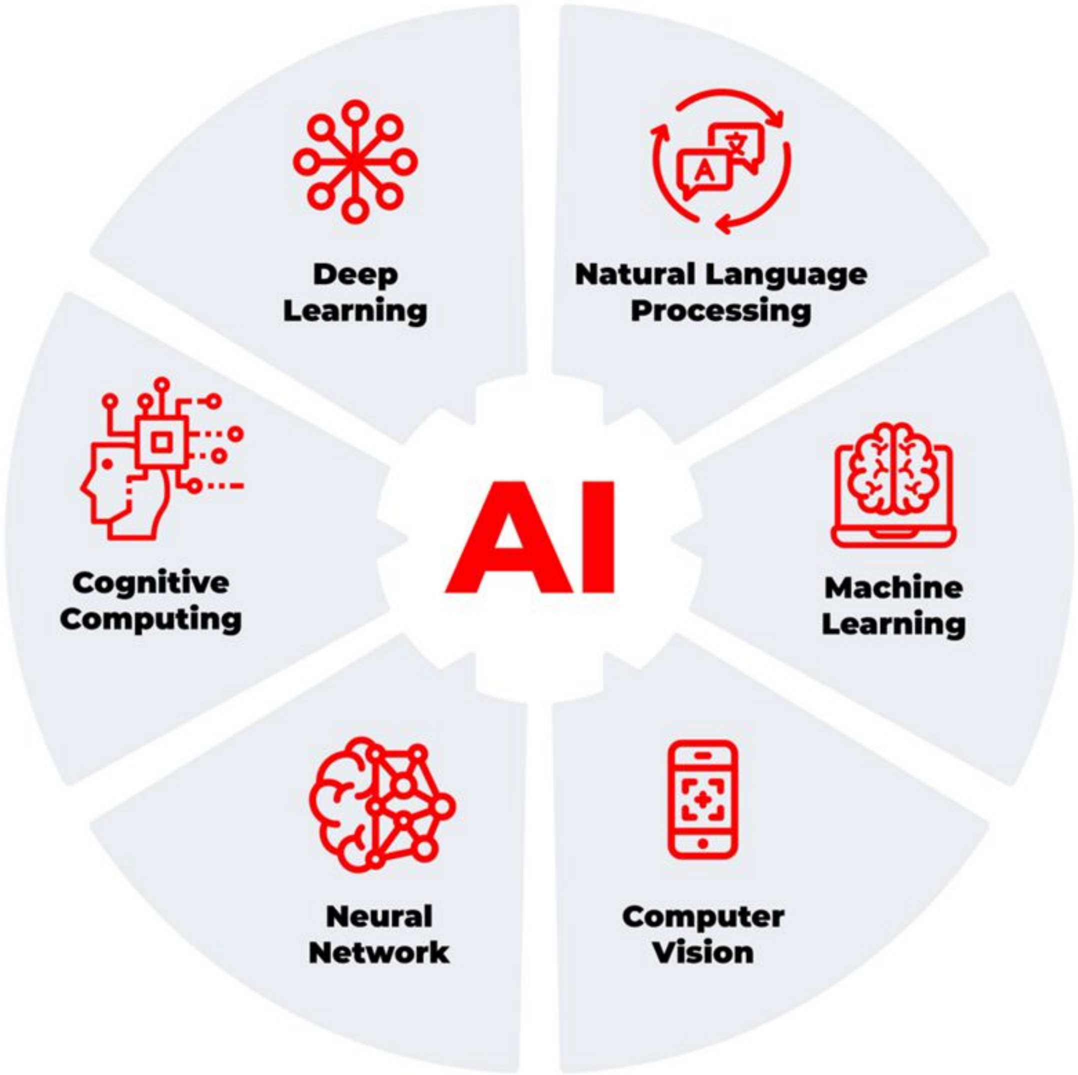
The Tower of Hanoi

Somewhere near Hanoi there is a monastery whose monks devote their lives to a very important task. In their courtyard are three tall posts. On these posts is a set of sixty-four disks, each with a hole in the center and each of a different radius. When the monastery was established, all of the disks were on one of the posts, each disk resting on the one just larger than it. The monks' task is to move all of the disks to one of the other pegs. Only one disk may be moved at a time, and all the other disks must be on one of the pegs. In addition, at no time during the process may a disk be placed on top of a smaller disk. The third peg can, of course, be used as a temporary resting place for the disks. What is the quickest way for the monks to accomplish their mission?

Even the best solution to this problem will take the monks a very long time. This is fortunate, since legend has it that the world will end when they have finished.

The Monkey and Bananas Problem

A hungry monkey finds himself in a room in which a bunch of bananas is hanging from the ceiling. The monkey, unfortunately, cannot reach the bananas. However, in the room there are also a chair and a stick. The ceiling is just the right height so that a monkey standing on a chair could knock the bananas down with the stick. The monkey knows how to move around, carry other things around, reach for the bananas, and wave a stick in the air. What is the best sequence of actions for the monkey to take to acquire lunch?



AI
Deep Learning
Natural Language Processing
Cognitive Computing
Machine Learning
Neural Network
Computer Vision

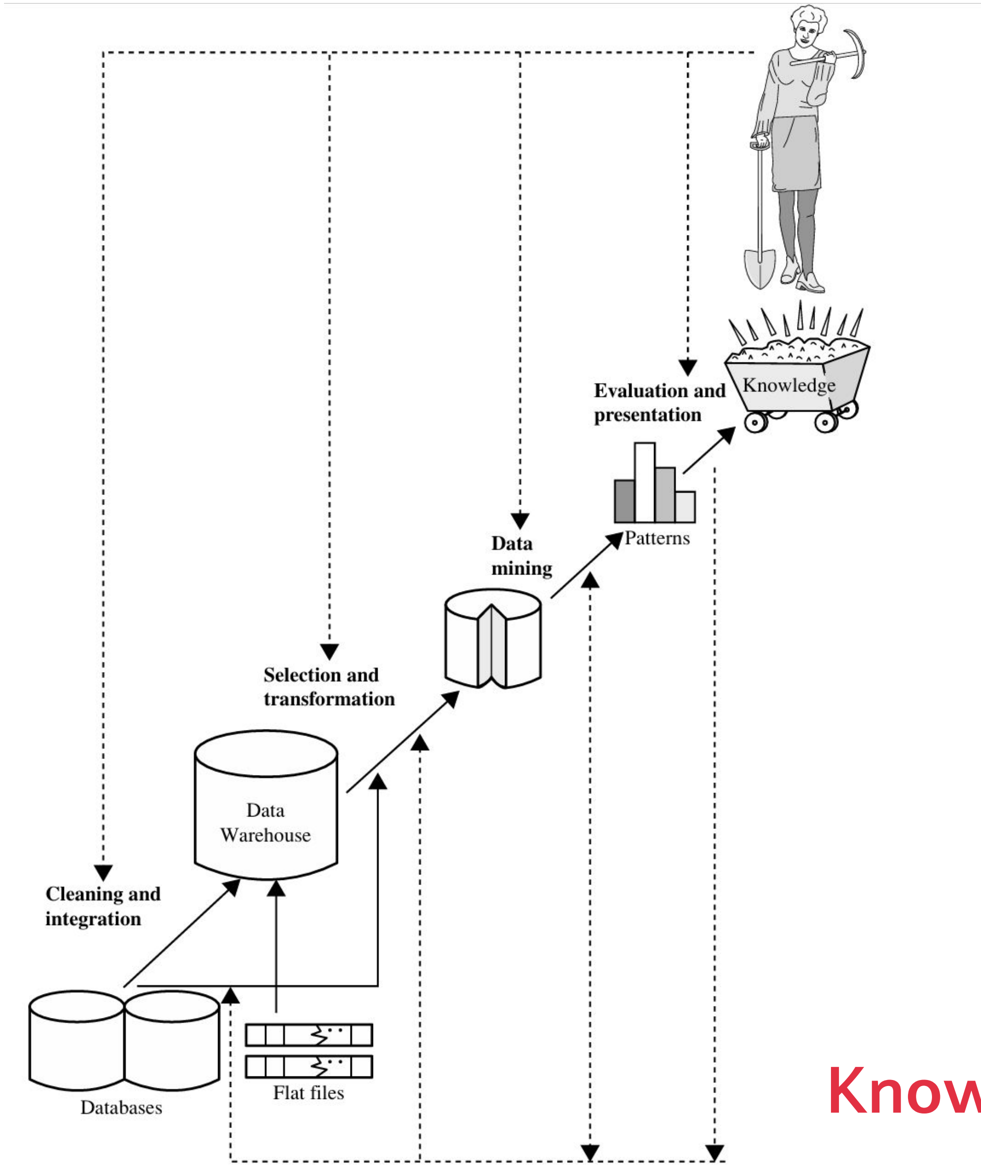
How do we define Data Mining?

Formal Definition

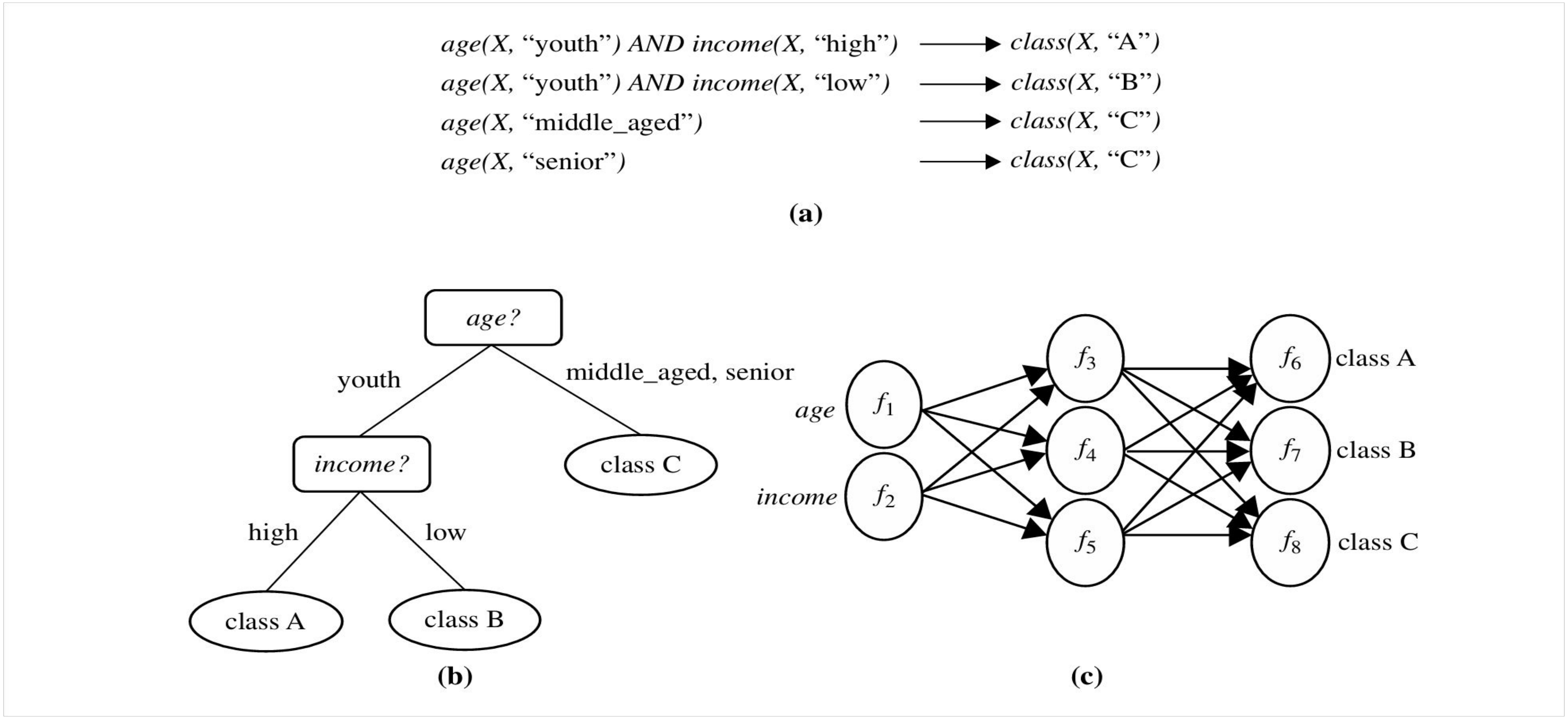
Data mining is the **process of discovering interesting patterns and knowledge from large amounts of data.** The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.

Process of Knowledge Discovery from Data

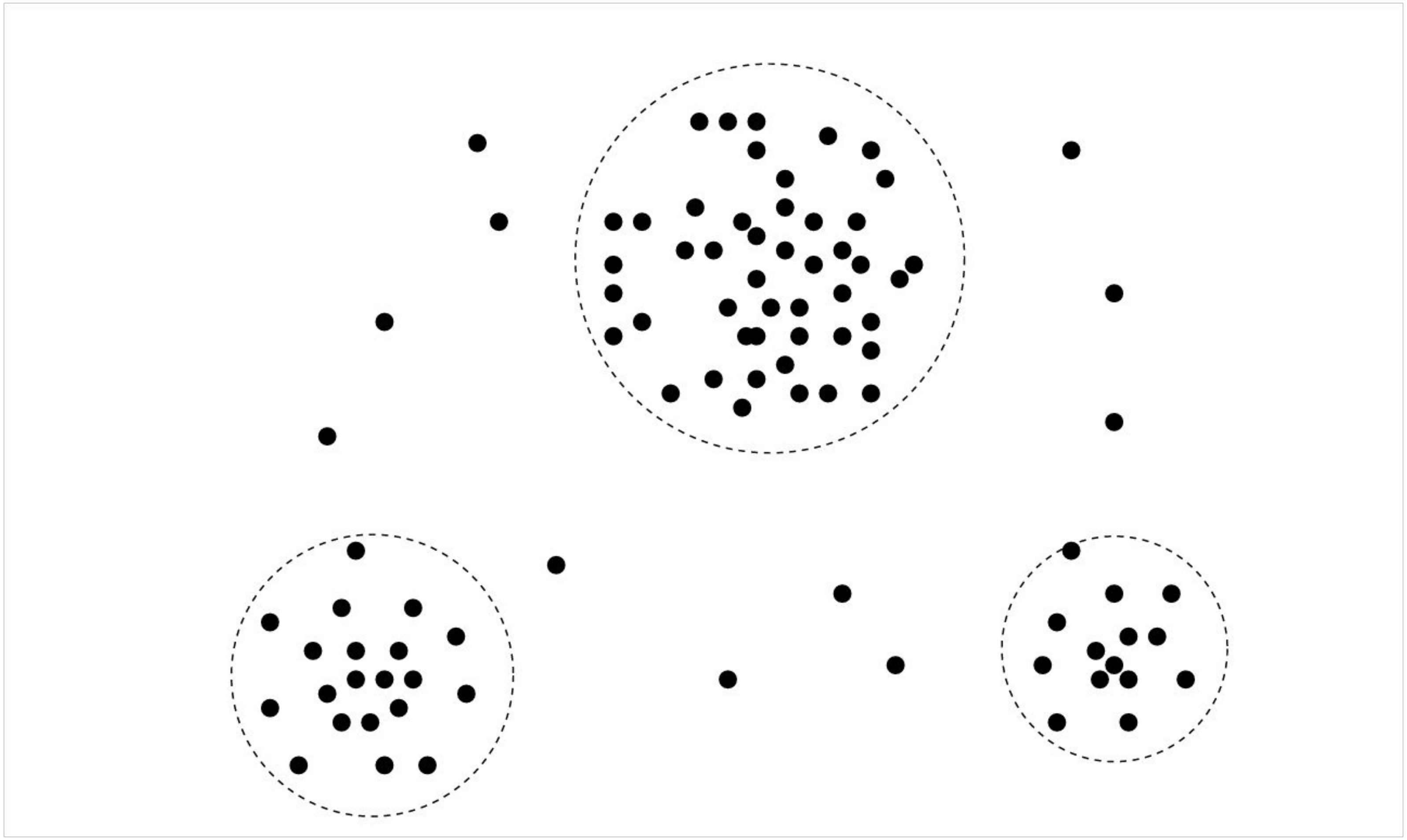
1. **Data Cleaning**: To remove noise and inconsistent data.
2. **Data Integration**: Where multiple data sources may be combined.
3. **Data selection**: Where data relevant to the analysis task are retrieved from the database.
4. **Data transformation**: Where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations.
5. **Data Mining:** an essential process where intelligent methods are applied to extract data patterns).
6. **Pattern Evaluation:** to identify the truly interesting patterns representing knowledge based on interestingness measures.
7. **Knowledge presentation:** where visualization and knowledge representation techniques are used to present mined knowledge to users.



General Process of
Knowledge Discovery from Data



Representation of Classification using different types of data modelling



Cluster Analysis of Random Data

Terms about Interestingness about Data

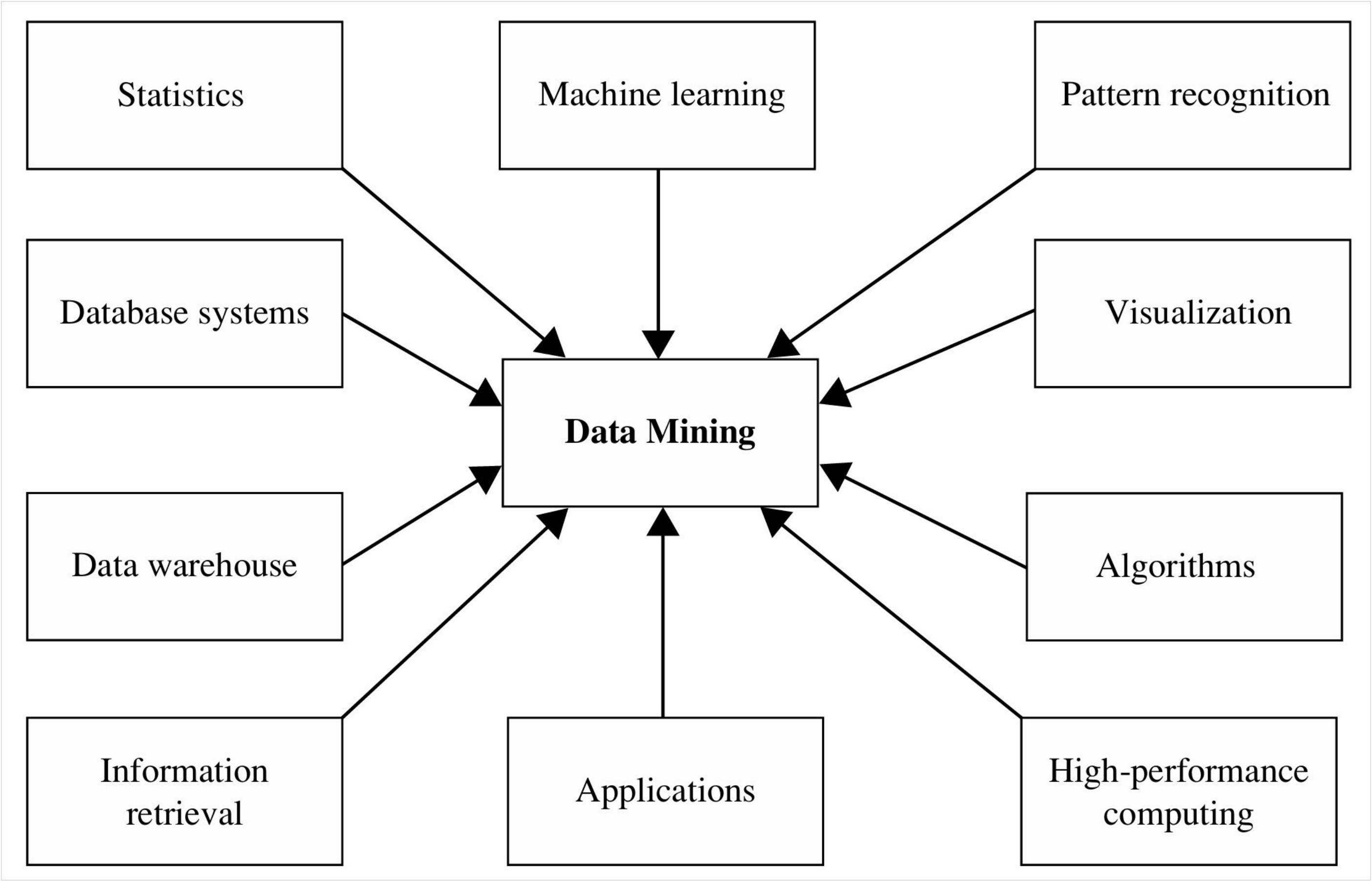
1. **Outlier Analysis**
   - A data set may contain **objects that do not comply with the general behavior or model** of the data. These data objects are **outliers**. Many data mining methods discard outliers as noise or exceptions. In some applications e.g., **fraud detection** the **rare events** can be **more interesting than the more regularly occurring ones**. The analysis of outlier data is termed as **outlier analysis** or **anomaly mining**.

2. **Interestingness of a Pattern in Data Mining**
   - Not all patterns are interesting, only the ones that have a particular threshold of **support and confidence** are interesting.

$$support(X \Rightarrow Y) = P(X \cup Y),$$

$$confidence(X \Rightarrow Y) = P(Y|X).$$

Data Mining is an Interdisciplinary Field



Disciplines of Data Mining

1. **Statistics**
   a. Focuses on data collection, analysis, and presentation.
   b. Helps build statistical models for data classification, pattern recognition, and prediction.
2. **Machine Learning**
   a. Enables computers to learn patterns and make decisions from data.
   b. Includes supervised (classification), unsupervised (clustering), and semi-supervised learning.
3. **Database Systems and Data Warehouses**
   a. Manages large-scale structured data efficiently.
   b. Offers scalability for large datasets, often through data warehouses.
   c. Enables data mining by integrating data from multiple sources.
4. **Information Retrieval**
   a. Searches for unstructured information, often using probabilistic models.
   b. Useful for text and multimedia mining, especially on large data collections like the Web.

Applications Targeted by Data Mining

1. **Business Intelligence (BI)**:
   a. BI helps businesses understand **customer behavior, market trends, competitors, and operations**.
   b. Core techniques include classification, prediction, clustering, and OLAP for sales, customer management, and market insights.
2. **Web Search Engines**
   a. Search engines use data mining to **handle large-scale web data**, including **crawling**, indexing, and ranking search results.
   b. Data mining improves query classification, personalized recommendations, and context-aware search results.

Issues in Data Mining

1. **Mining Methodology** – Develop new methods to handle diverse knowledge types, data uncertainty, and multidimensional data mining.
2. **User Interaction** – Enhance interactive mining, incorporate user knowledge, and improve result visualization.
3. **Efficiency and Scalability** – Ensure mining algorithms are scalable, efficient, and capable of handling large, distributed data.
4. **Diversity of Data Types** – Address the challenges of mining structured, semi-structured, and unstructured data across diverse sources.
5. **Data Mining and Society** – Focus on social impact, privacy concerns, and seamless integration of mining into daily life.

Types of Data Attributes

An **attribute** is a **data field** representing a characteristic or feature of a **data object**.

1. Nominal Attributes are **categorical data** with values that represent names or categories like hair color or marital status. **No meaningful order, mathematical operations don't apply.**

2. Special case of nominal data with **only two possible values** is **Binary Data**, like true/false. These can be symmetric both states are **equally important** (gender) or asymmetric **one state is more important** (cancer).

3. **Ordinal Values** have ordered values such as sizes like small, medium, large, but the difference between successive values is unknown. They are useful for ranking but not for calculating averages.

4. **Quantitative attributes** with purely measurable values either interval-scaled or ratio-scaled. Allow all mathematical operations.

Advantages of Data Mining

1. Data mining **excels at identifying patterns and trends within large datasets**, which can lead to valuable insights for decision-making processes across various sectors such as healthcare, finance, and marketing.
2. Enables organizations to predict **future trends based on historical data**. For instance, businesses can forecast customer purchasing behaviors, allowing for targeted marketing strategies (Market Basket Analysis).
3. By automating the **data analysis process** data mining enhances efficiency.
4. Companies can gain a deeper understanding of customer preferences and behaviors enabling personalized marketing efforts that **improve customer satisfaction and loyalty**.

Disadvantages of Data Mining

1. The effectiveness of data mining is highly dependent on the **quality of the data** used. Poor quality or biased data can lead to inaccurate predictions and misleading insights.
2. The use of personal data raises ethical issues regarding privacy. Organizations must **navigate legal frameworks and ethical considerations** when mining sensitive information.
3. There is a **risk of overfitting** models to training data.
4. Implementing data mining techniques requires **significant expertise in statistics and machine learning.**

Resources

- **Kevin Knight, Elaine Rich, B. Nair – Artificial Intelligence** – Tata McGraw-Hill Education Pvt. Ltd. (2010)
- **Jiawei Han, Jian Pei, Hanghang Tong – Data Mining Concepts and Techniques** – The Morgan Kaufmann Series in Data Management Systems
- **Russell S , Norvig P – Artificial Intelligence – A Modern Approach,** Second Edition

How do we define Machine Learning?

Formal Definition

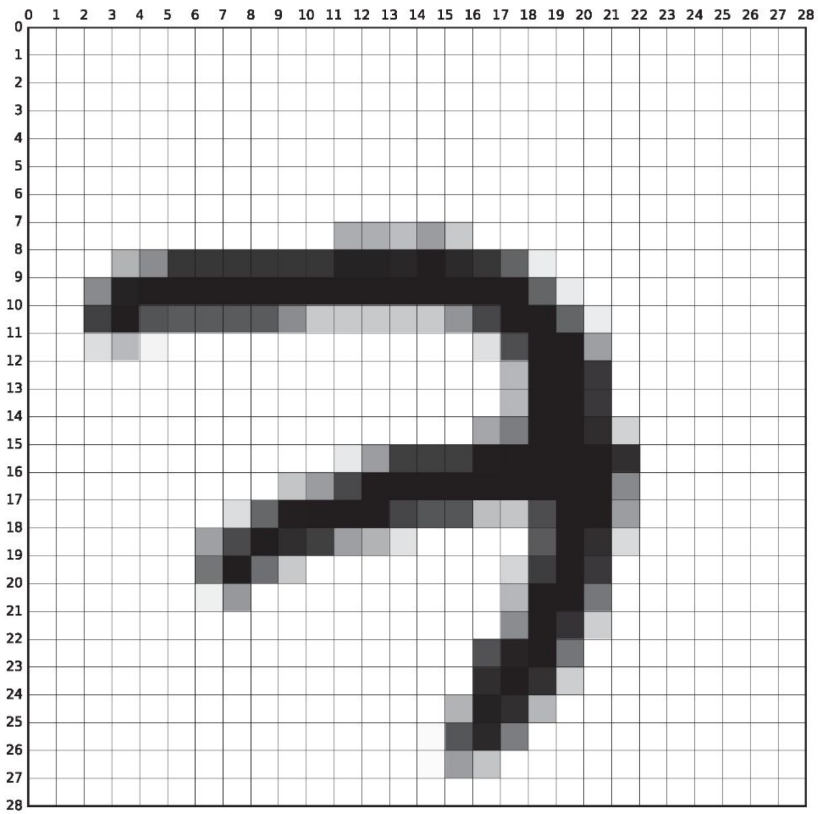
Machine Learning is a branch of artificial intelligence and computer science that focuses on the using data and algorithms to enable AI to enact the way that humans learn **gradually improving its accuracy**.

Book Definition

Machine learning investigates how computers can learn (or improve their performance) based on data. A main research area is for computer programs to learn to recognize complex patterns and make intelligent decisions based on data.

For example, a typical machine learning problem is to program a computer so that it can automatically recognize handwritten postal codes on mail after learning from a set of examples.

MNIST Numerical Character Dataset



(**a**) MNIST sample belonging to the digit ‘7’.

(**b**) 100 samples from the MNIST training set.

Simple Analogy

A child to **recognize different types of fruits**. Instead of giving them a detailed description of each fruit, you show them various examples.

1. **Initial Exposure**
   - Present the child with pictures of apples, bananas, and oranges. Point out the colors, shapes, and sizes.
2. **Learning Through Examples**
   - Overtime sees many apples, bananas, and oranges. They start to notice patterns apples are usually **round and red or green**, bananas are **long and yellow**, and oranges are **round and orange**.
3. **Testing Knowledge**:
   - A new fruit they haven't seen before a pear and ask them what it is. Based on their previous experiences, **they might say it looks like an apple but is not quite the same.**
4. **Refinement**
   - Correct them and explain why it's different. The next time they see a pear, they'll remember the distinction.

Types of Machine Learning

Based on the **type data we provide** and **type of algorithm employed** while learning, there are **3 broad categories**

1. **Supervised Learning**
   - The learning is done based on a **labeled dataset -** meaning that each training example is has a corresponding output label. Ex: Classification, Regression

2. **Unsupervised Learning**
   - Involves learning on data **without labeled outputs**. The algorithm/process/computer tries to identify **patterns and relationships** in the data on its own. Ex: Clustering

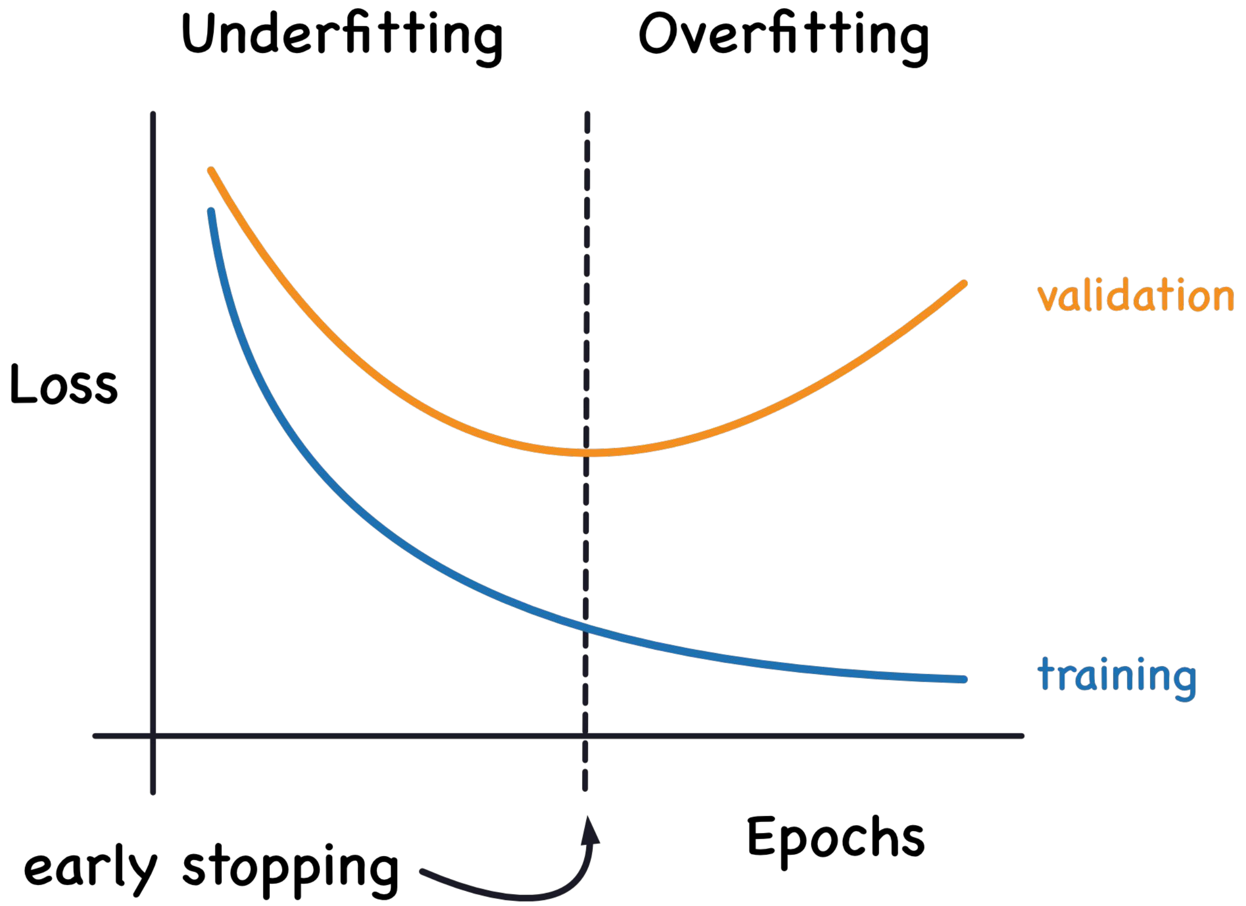
3. **Reinforcement Learning**
   - Learning is based on making decisions by **taking actions in an environment** to **maximize cumulative reward**. The agent **receives feedback** in the form of **rewards or penalties** based on its actions. Ex: A robot trying to map the environment based on feedbacks.

Underfitting vs Overfitting

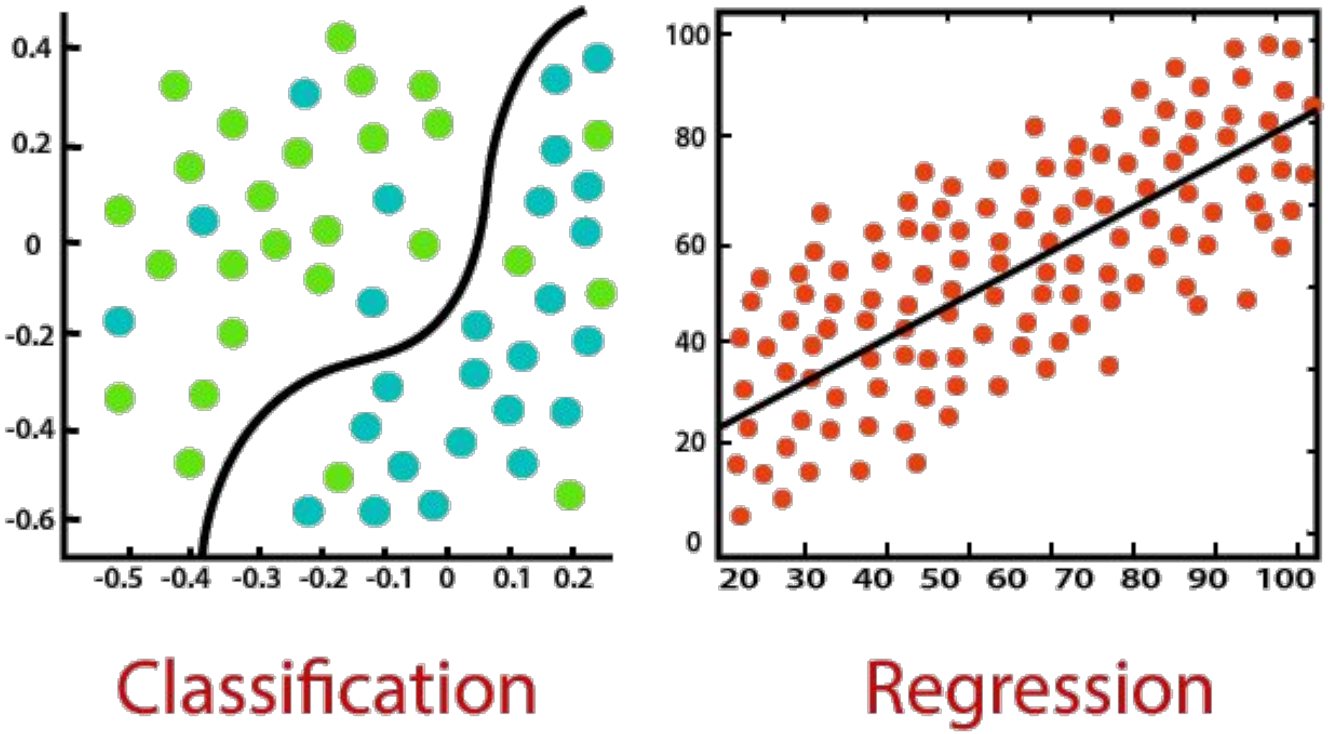
Based on the **amount of the data the model has generalized or learned or trained on, the learning level is defined in 2 broad classes**

1. **Underfitting**
   - Occurs when a model is too simple to capture the underlying structure of the data.

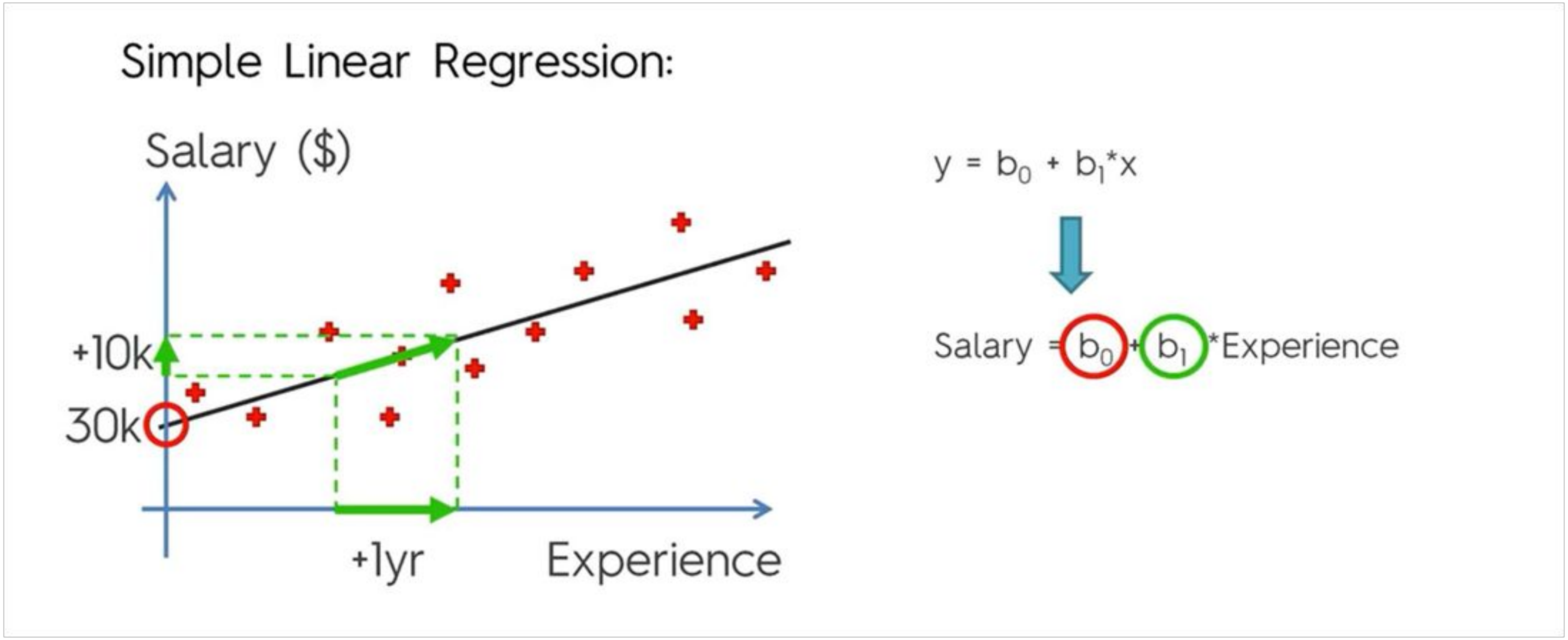
2. **Overfitting**
   - Occurs when a model is too complex and captures not just the underlying structure but also the noise or random fluctuations in the training data.

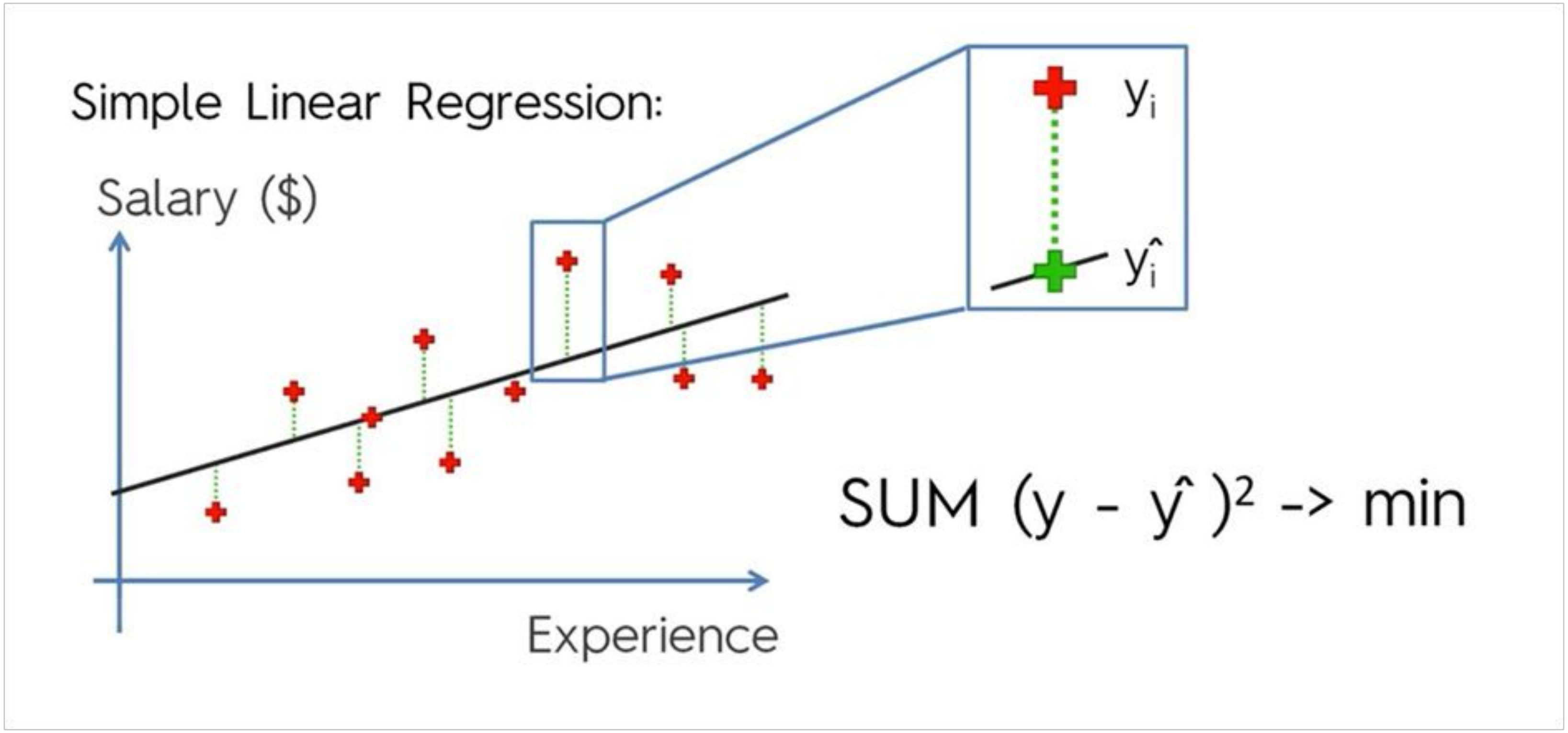

Underfitting
Overfitting
Loss
validation
training
Epochs
early stopping

Classification vs. Regression

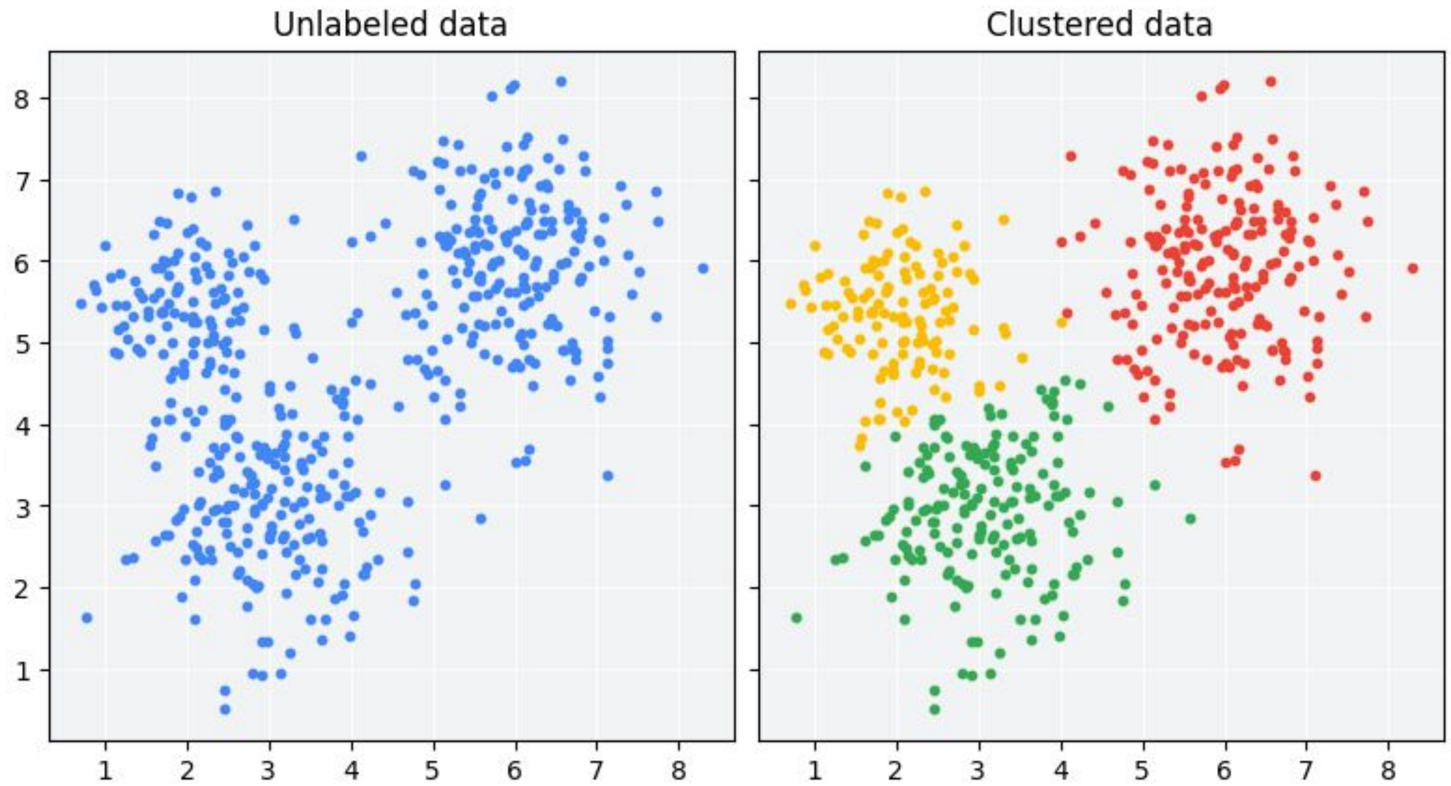


Simple Regression



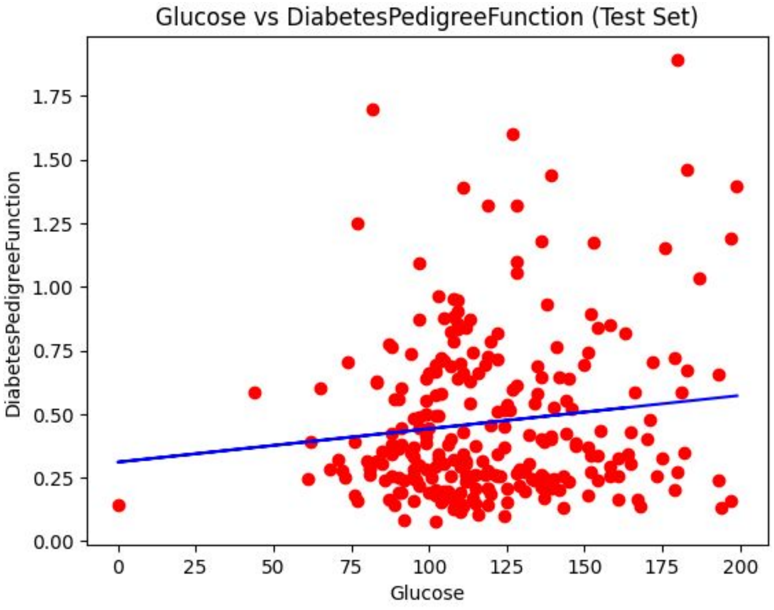

Simple Linear Regression:
Salary ($)
Experience
y_i
$\hat{y}_i$
SUM $(y - \hat{y})^2$ -> min

Clustering



Sample ML Program

https://www.kaggle.com/code/hardikgupta0808/lab-2-simple-linear-regression



Questions & Answers

That's all folks!

How do we define
Deep Learning?

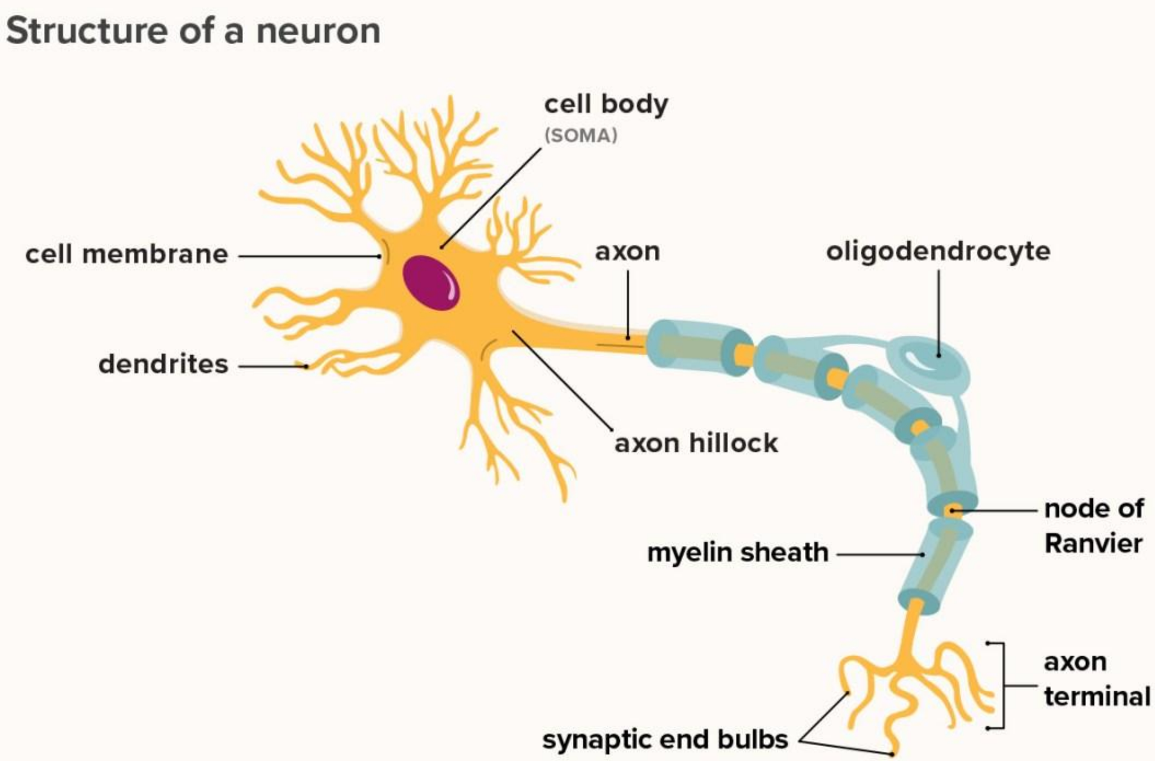
Formal Definition

A machine learning algorithm is an algorithm that is able to learn from data. But what do we mean by learning? Mitchell (1997) provides a succinct definition: “A computer program is said to learn from experience E with respect to someclass of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.”

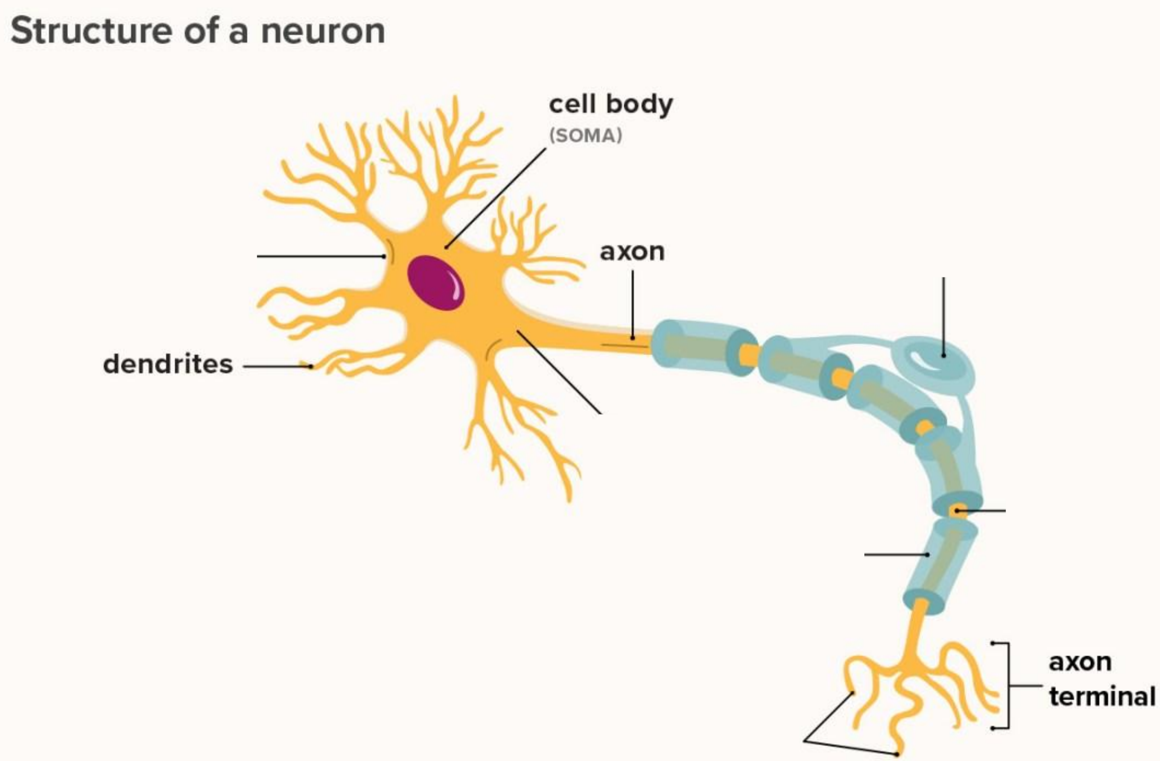
Formal Definition

Deep learning is a specialized subset of machine learning that utilizes neural networks to process data in a manner inspired by the human brain.

Neural Network!?

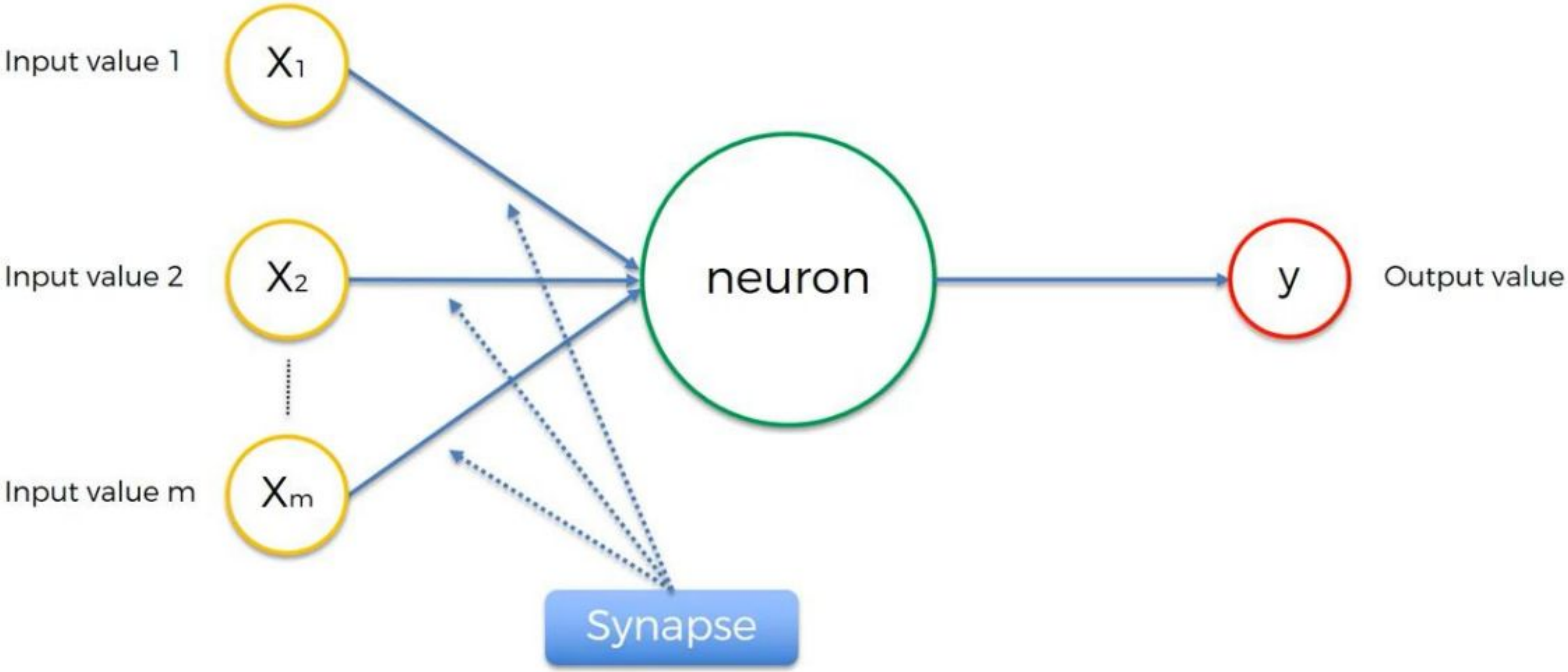


Neural Network!?



Break it down to The Neuron.

The smallest Mathematical Unit of any neural network.



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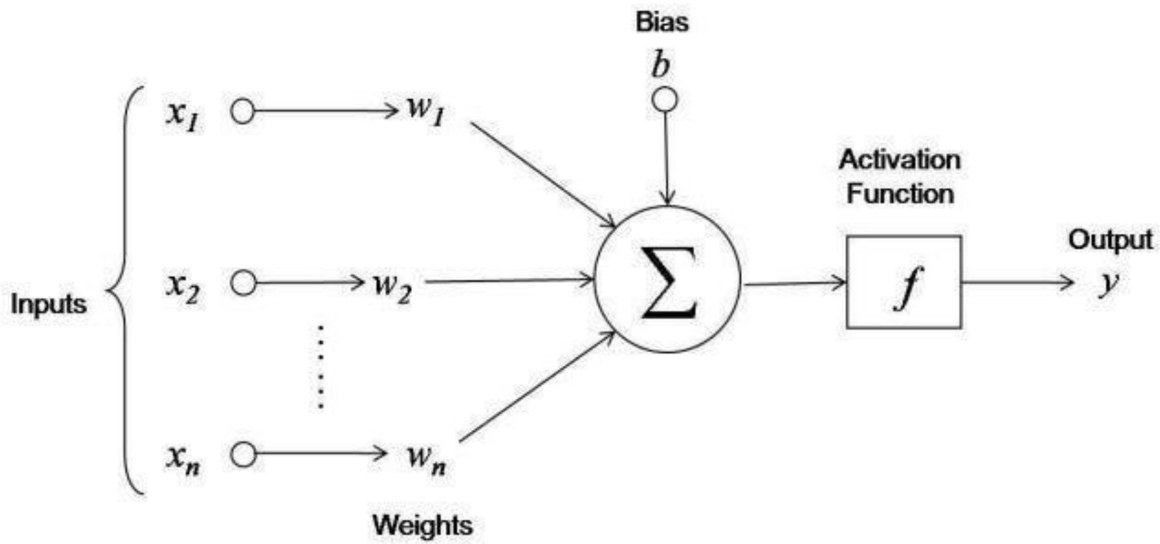
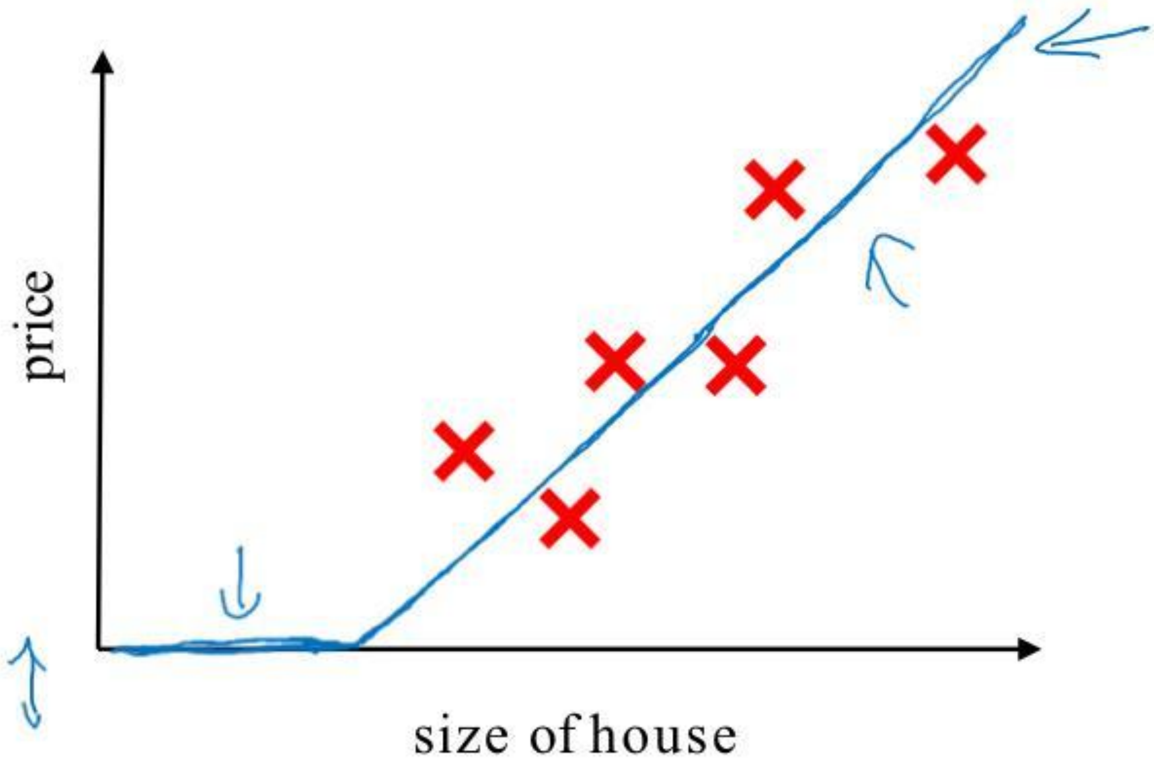
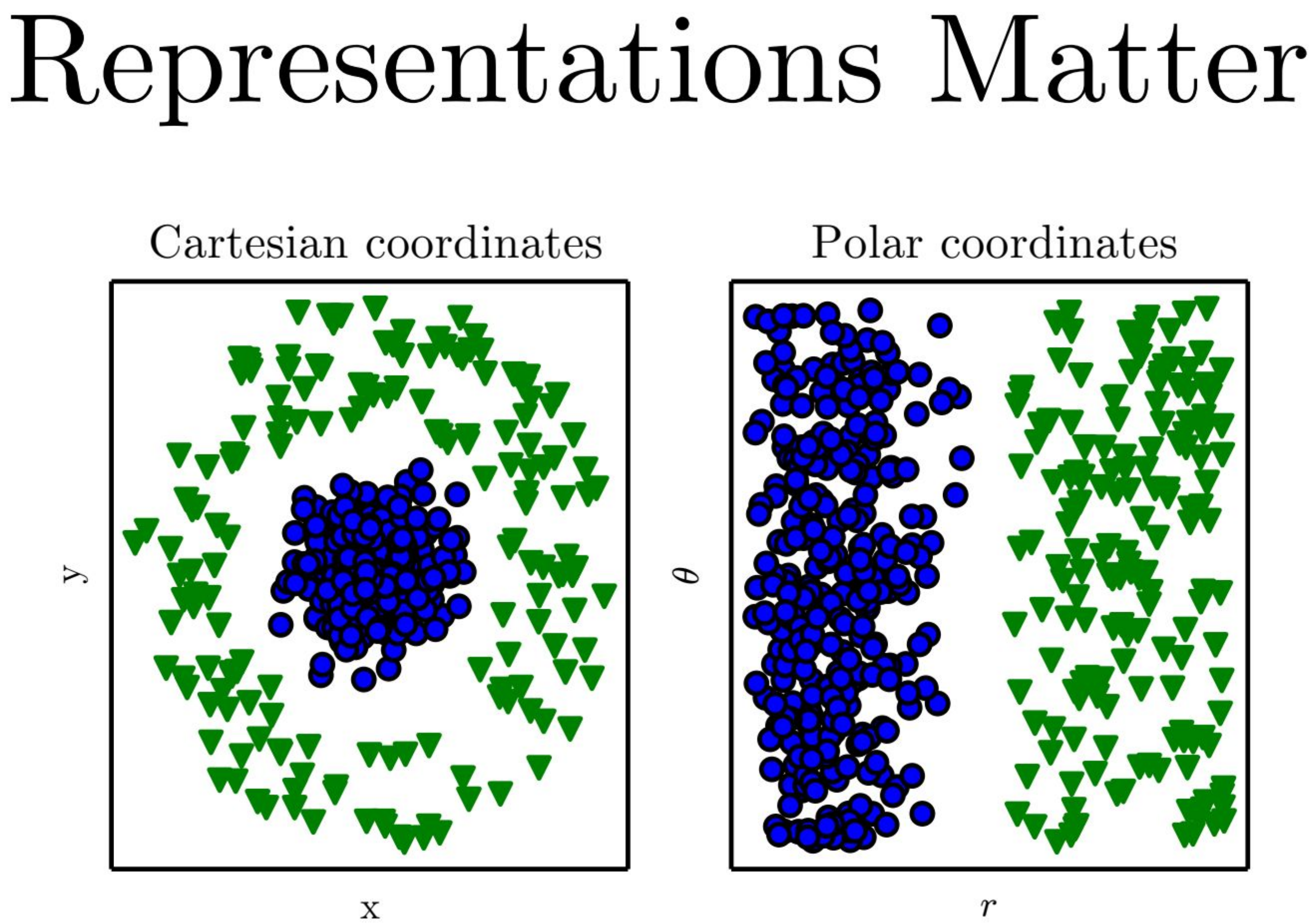


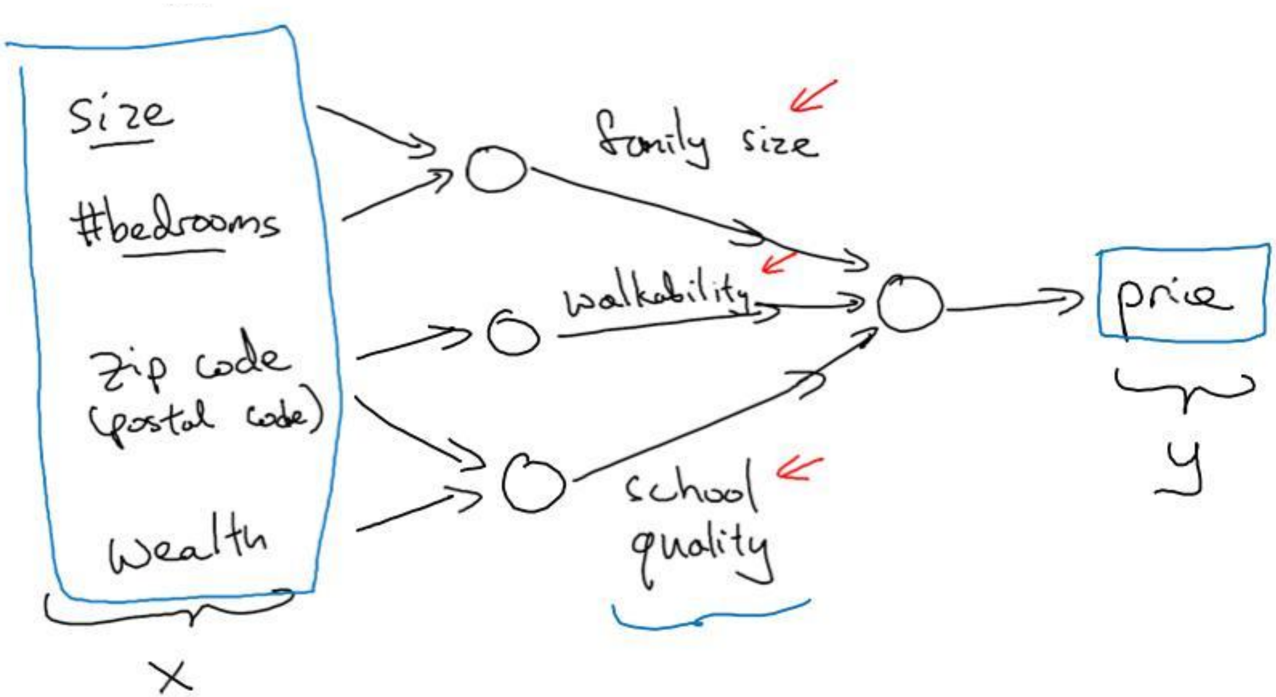
House Price Prediction



Representation of Data!?



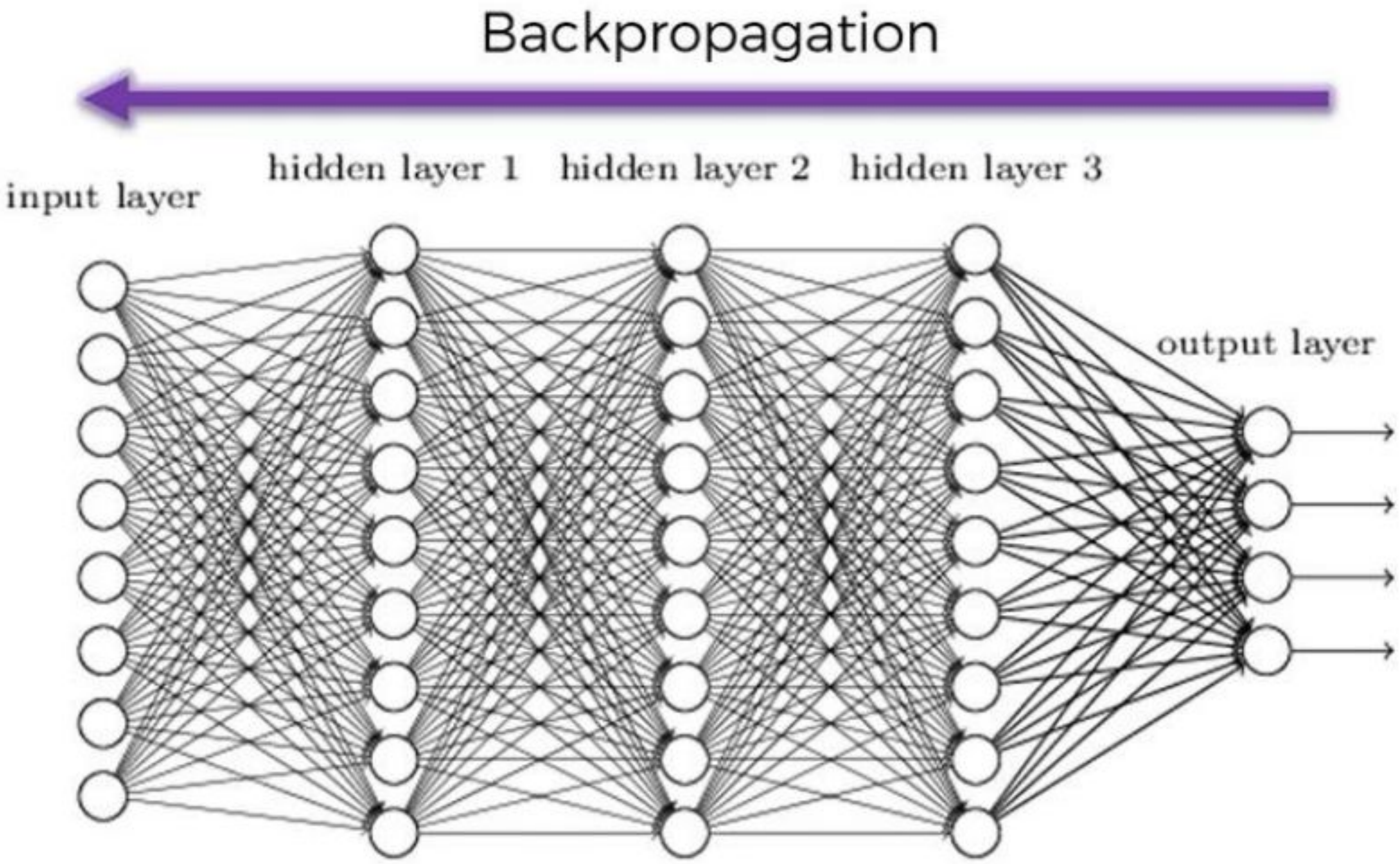
House Price Prediction using Neural Network.



House Price Prediction using Neural Network.

- Keep input nodes equal to number of independent variable
- Assign random weight to synapsis (links) – usually start with random number between -0.5 to 0.5
- Assign random biases – – usually start with random number between -0.5 to 0.5
- Calculate output $Y_p = Y_{<\text{predicted}>}$
- Calculate error $Y_a = Y_{<\text{actual}>} - Y_p$

Backpropagation in Neural Network.



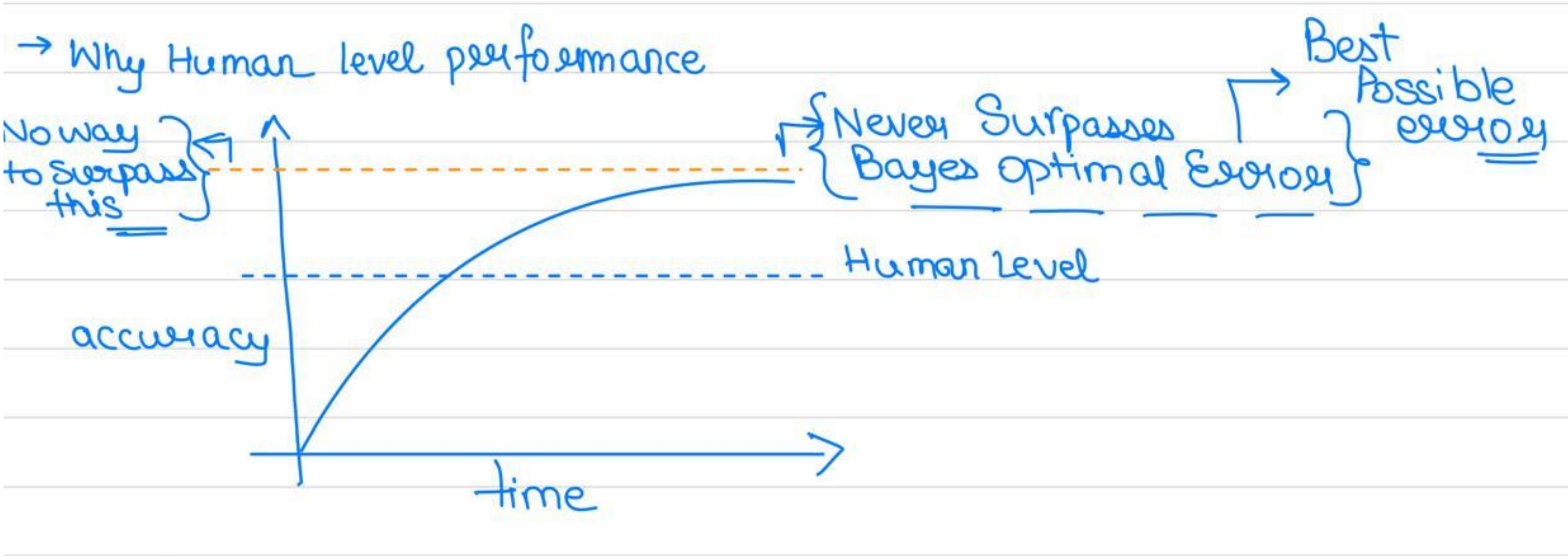
Why it works

- Because errors are used to adjust the weight
- Adjustment is proportionate to the error
  - Big errors lead to big changes in weights
  - Small errors leave weights relatively unchanged
- Over many such updates, a given weight keeps changing until the error associated with that weight is negligible
- At this point weights change little

Stopping Criteria

- When weights and bias hardly changes
- When errors are below a threshold
- When maximum number of iterations has reached

Performance of The Neural Network.



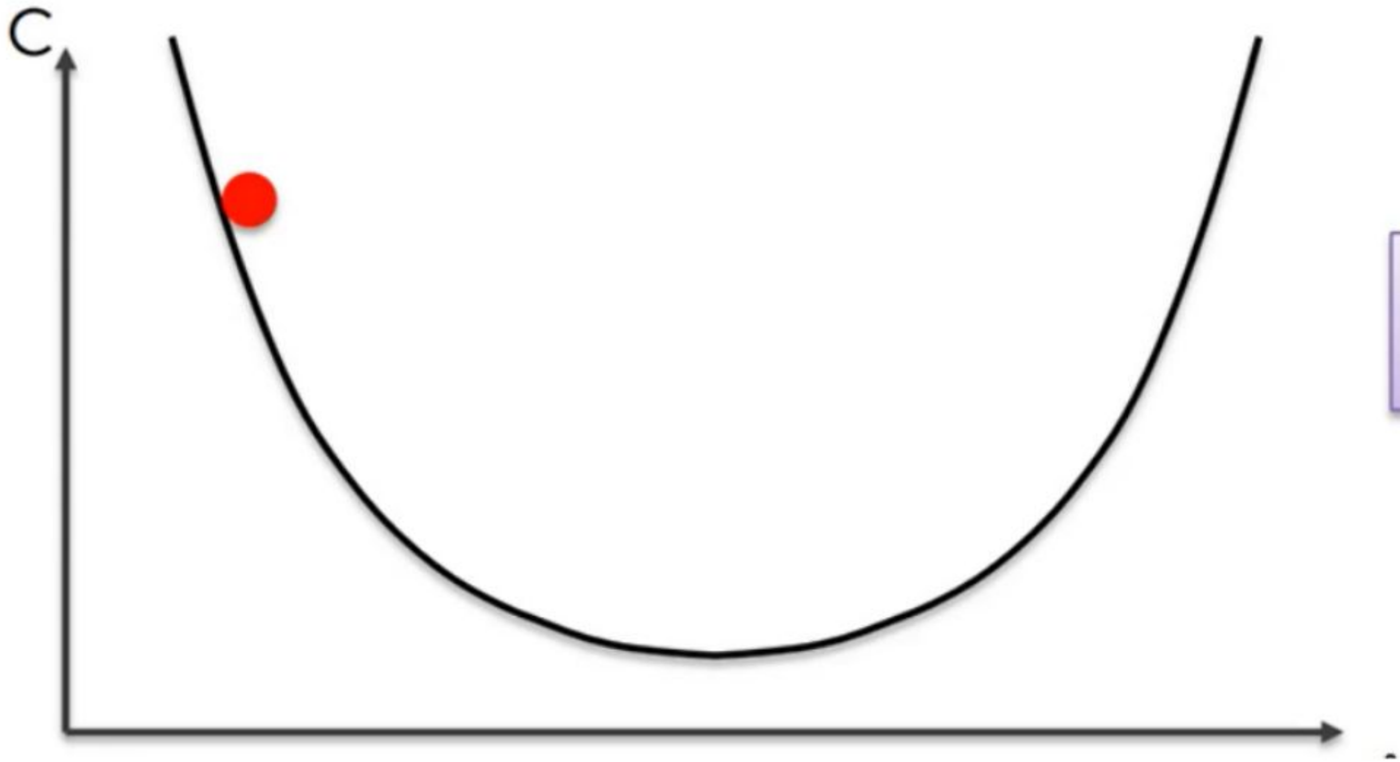
Back to Deep Learning

Utilizes **artificial neural networks** to process data in a **manner inspired by the human brain**.

Deep learning is a type of **machine learning** that employs **multilayered neural networks**, known as **deep neural networks**, to **automatically learn and extract features from large amounts of data.**

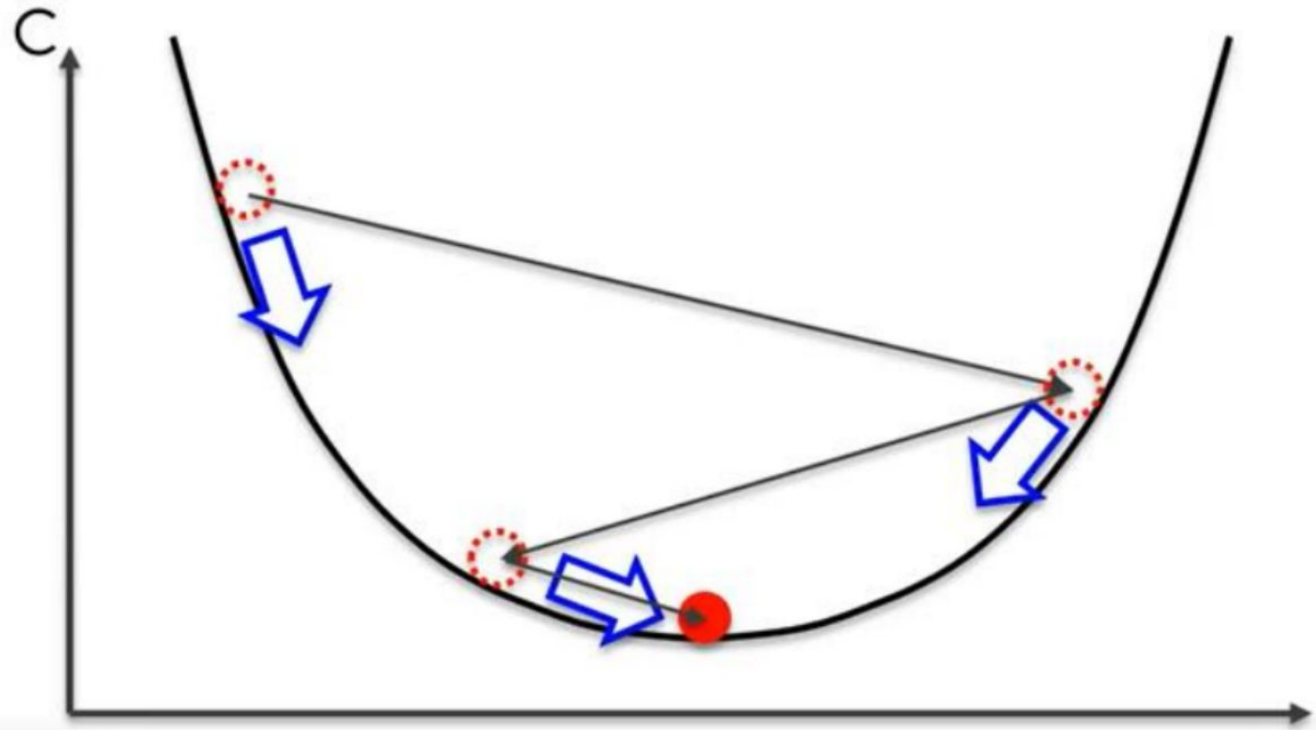
These networks consist of **numerous layers** of **interconnected neurons**, allowing them to recognize complex patterns in various forms of data, including **images**, **text**, and **audio**.

Error Minimization (Gradient Descent)



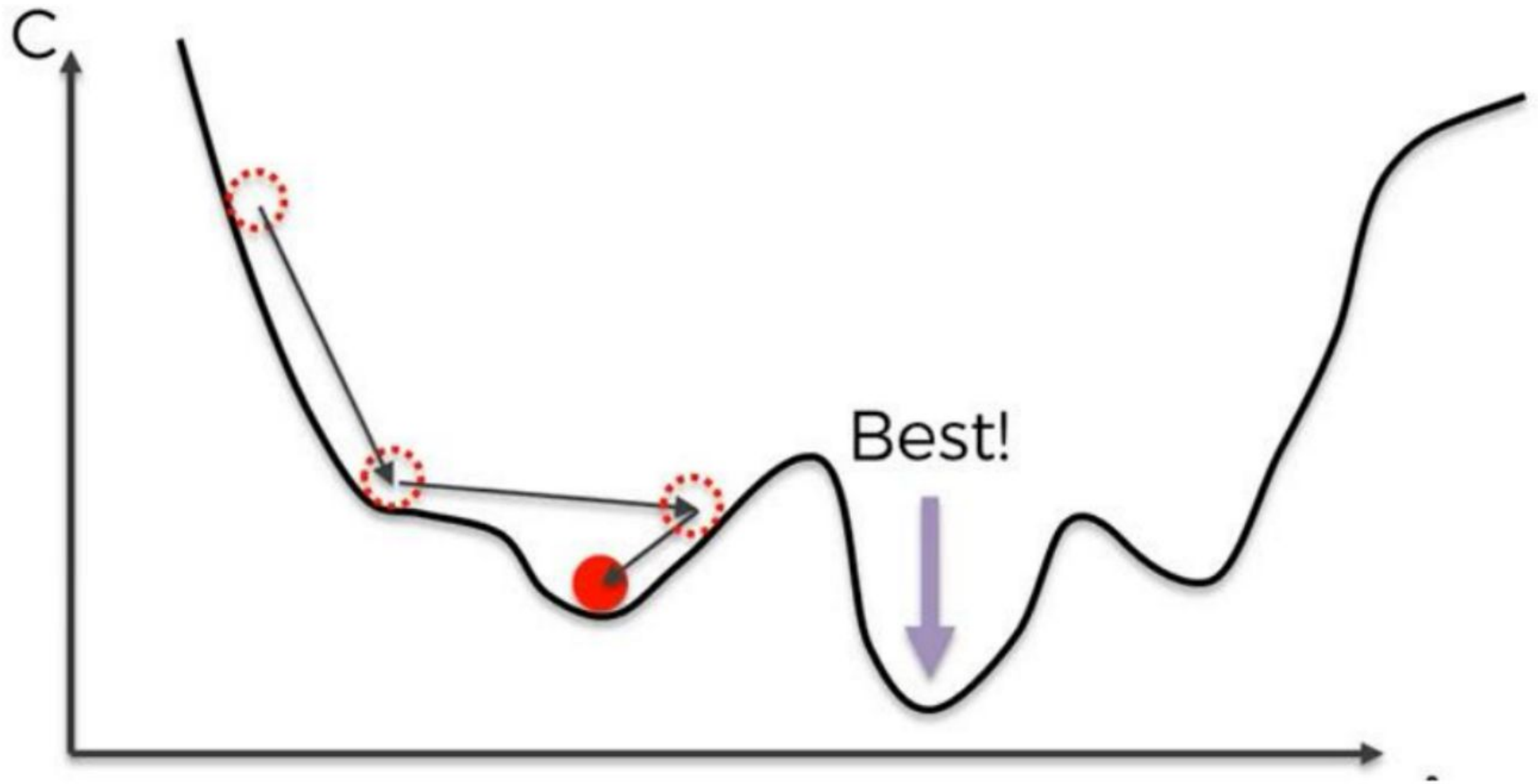
$$C = \frac{1}{2}(\hat{y} - y)^2$$

Error Minimization (Gradient Descent)



$$C = \frac{1}{2}(\hat{y} - y)^2$$

Local Minima in Error Minimization



Confused about **how?**

- **How** an image is used in neural network?
- **How** the text is used in neural network for NLP?
- **How** the audio is used in neural networks?

How do we define Data Science?

Formal Definition

An **interdisciplinary field** that focuses on extracting knowledge and insights from structured and unstructured data using scientific methods, algorithms.

These insights are used to **guide decision making** and **strategic planning**.

DATA + SCIENCE

refers to **raw facts and figures** that can be **collected and stored**

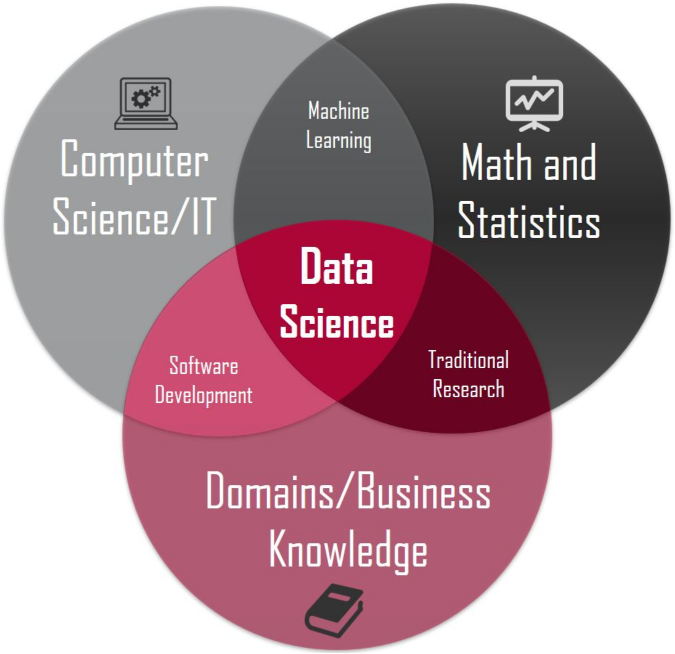
structured, unstructured

study of the **structure and behavior** of the physical and natural world through **observation and experimentation**

hypotheses, analyze results, and draw conclusions

Data Science Prerequisites

... interdisciplinary field?



Simple Analogy

Detective trying to solve a mystery:

- The detective collects **clues** (data)
- Interviews witnesses (gathers information or **domain knowledge**)
- and analyzes evidence (data analysis) to **solve the case** (get insights).

Real Life Example

Netflix recommendations

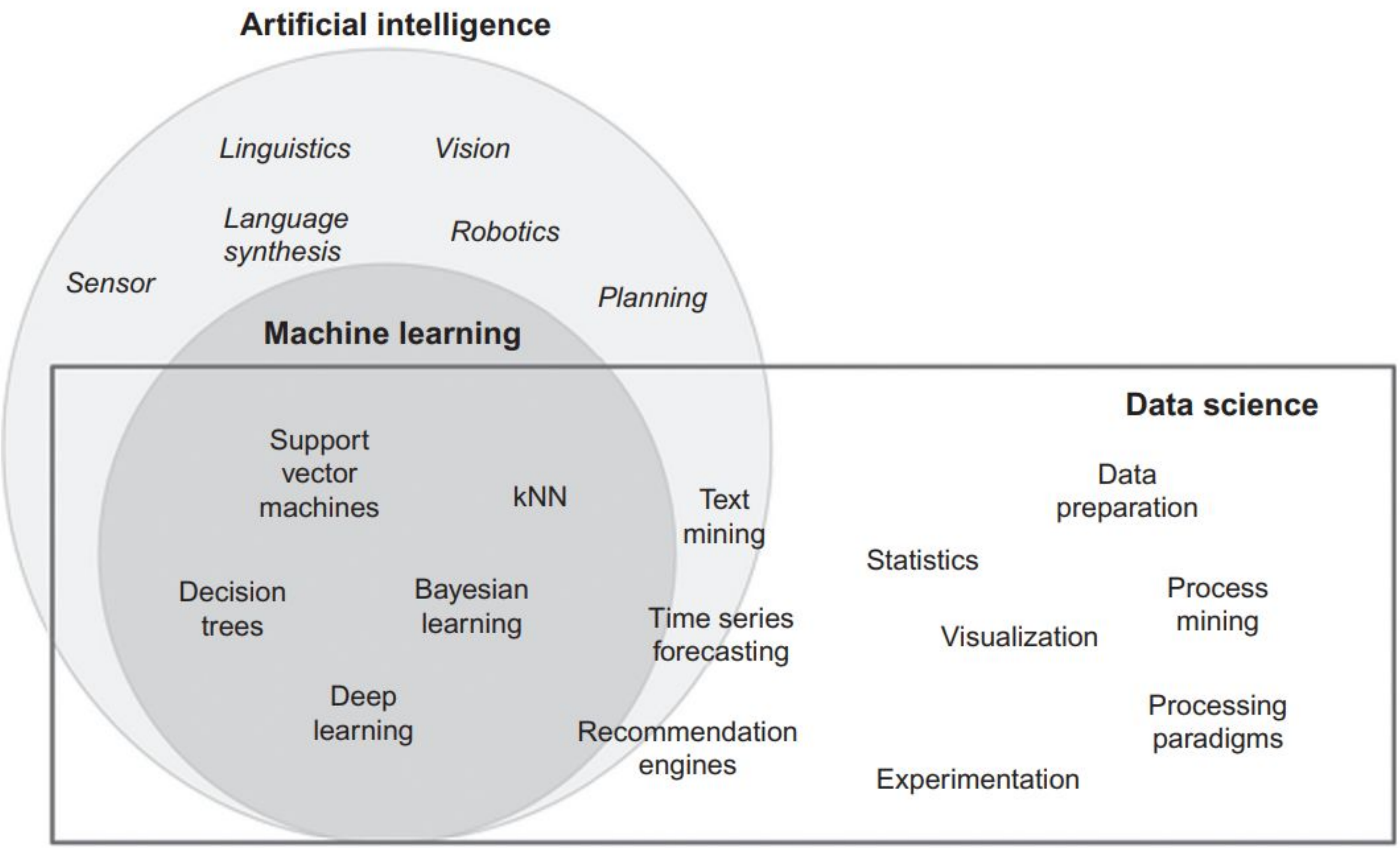
- Already has your **data, like what shows you watch, recent watch history and some personal data.** [Data ingestion]

- Already has the [Domain Knowledge]: **to recommend what you love to watch.**

- By identifying **patterns**, data scientists create algorithms that suggest content you are likely to enjoy **[Data Analysis]**

- [Communicate], the **gathered insights**, by recommending **particular shows**.

Relation with AI and ML

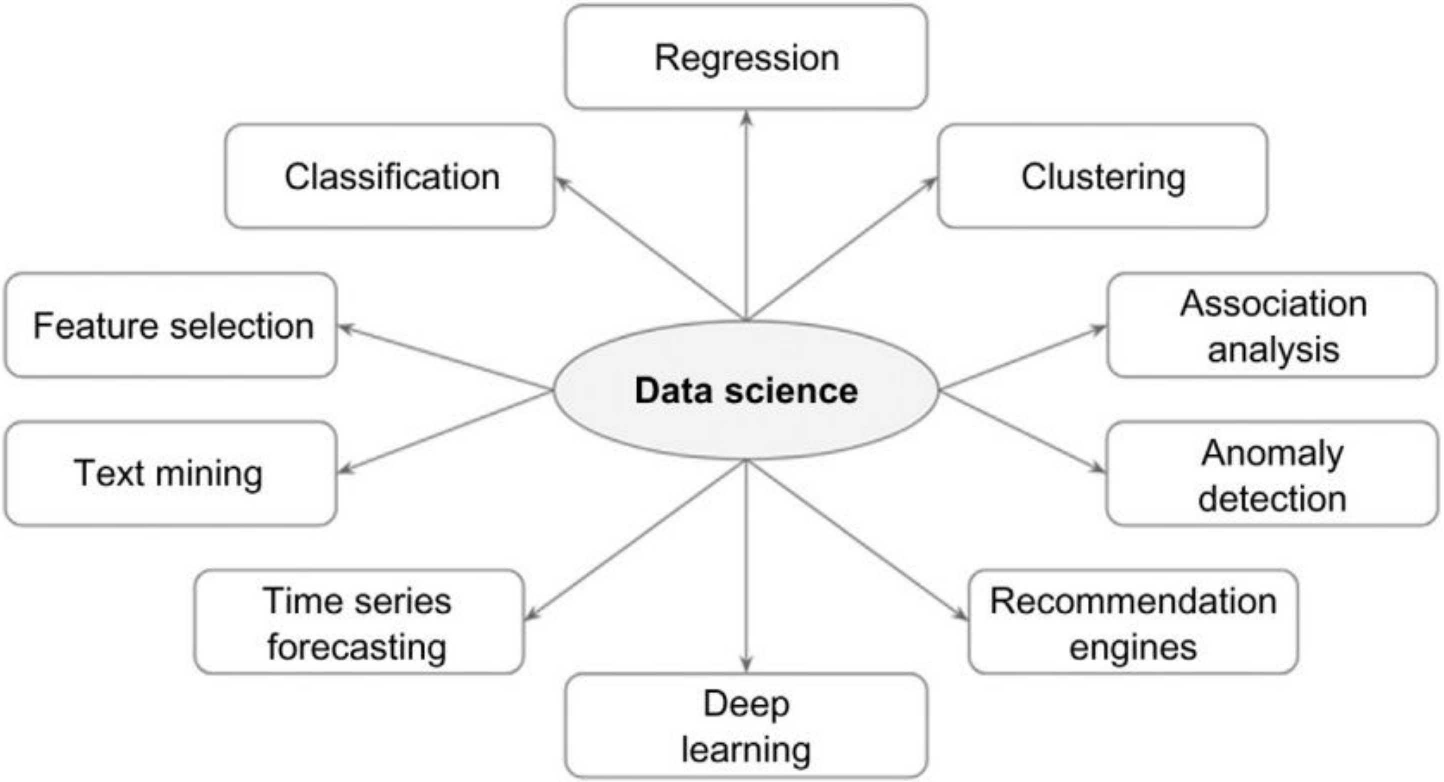


Fields Associated with Data Science

The techniques used in the steps of a data science process and in conjunction with the term “data science” are:

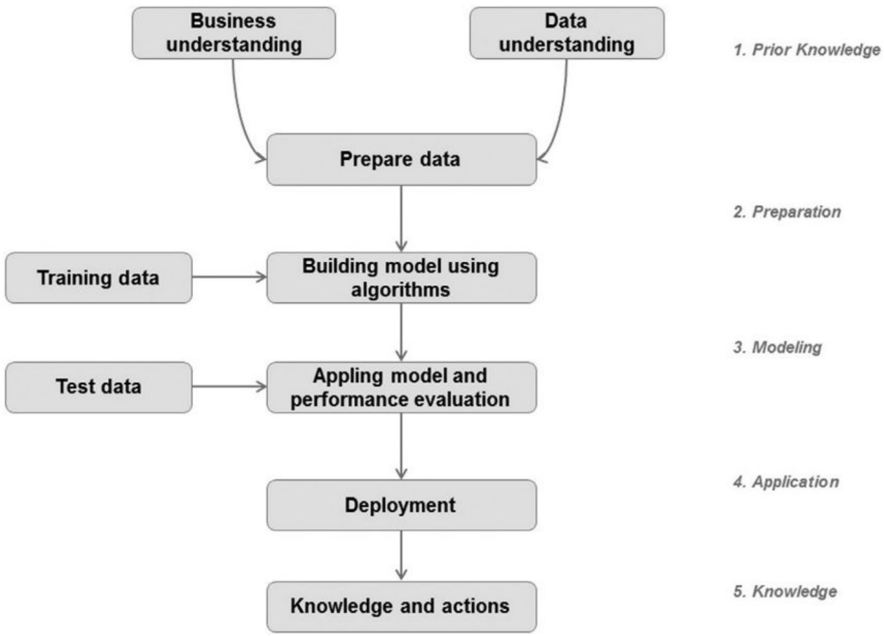
- **Descriptive statistics:** Computing mean, standard deviation, correlation, and other descriptive statistics, quantify the aggregate structure of a dataset.
- **Exploratory visualization:**The process of expressing data in visual coordinates enables users to find patterns and relationships in the data and to comprehend large datasets.
- **Hypothesis testing:** In confirmatory data analysis, experimental data are collected to evaluate whether a hypothesis has enough evidence to be supported or not. There are many types of statistical testing and they have a wide variety of business applications (e.g., A/B testing in marketing). In general, data science is a process where many hypotheses are generated and tested based on observational data.
- **Data engineering:** Data engineering is the process of sourcing, organizing, assembling, storing, and distributing data for effective analysis and usage. Database engineering, distributed storage, and computing frameworks (e.g., Apache Hadoop, Spark, Kafka), parallel computing, extraction transformation and loading processing, and data warehousing constitute data engineering techniques. Data engineering helps source and prepare for data science learning algorithms.

Data Science General Tasks

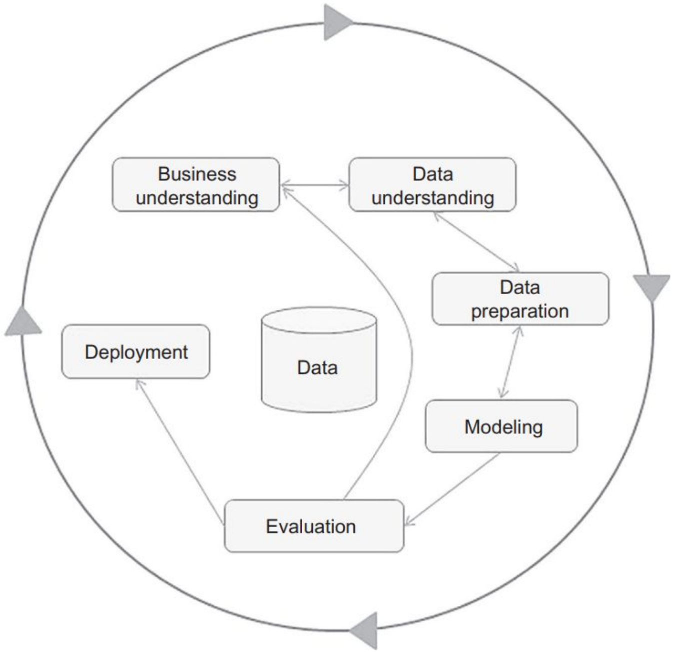


Standard Data Science Process

1. Understanding the Problem.
2. Preparing the data samples.
3. Developing the model.
4. Apply the model in dataset.
5. Deploying and maintaining the models.



One of the most popular data science process frameworks is **Cross Industry Standard Process for Data Mining (CRISP-DM)**.This framework was developed by a consortium of companies involved in data mining.The CRISP-DM process is the most widely adopted framework for developing data science solutions.



How do we define Cybersecurity?

Formal Definition

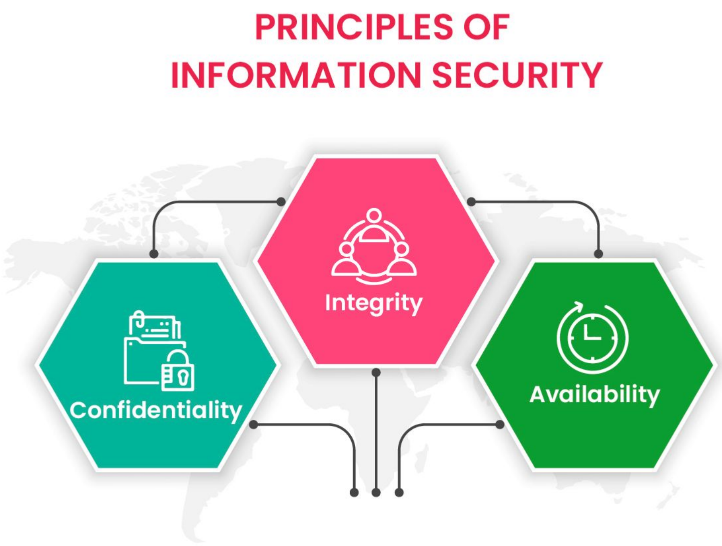
The practice of protecting systems, networks, and programs from digital attacks is known as Cybersecurity. It is used to safeguard sensitive information, prevent data breaches, and maintain the integrity and confidentiality of data.

Key Concepts of Information Security:

Confidentiality: Ensuring that information is only accessible to those authorized to access it.

Integrity: Protecting data from being altered without authorization.

Availability: Ensuring reliable access to information and resources when needed.



Cybercrime

- **Definition**: Illegal activities that involve a computer or network.
- **Types of Cybercrime**:
    - **Phishing**: Fraudulent attempts to obtain sensitive information.
    - **Ransomware**: Malware that encrypts files until a ransom is paid.
    - **Hacking**: Unauthorized access to data and systems.
- **Impact**: Financial loss, identity theft, damage to reputation.

Cybercrime

Aspect	Phishing	Ransomware	Hacking
Definition	Fraudulent attempts to obtain sensitive information.	Malware that encrypts files until a ransom is paid.	Unauthorized access to data and systems.
Method of Attack	Deceptive communication (emails, messages) to trick users.	Infects devices via malicious software, often through downloads.	Exploiting vulnerabilities in software or systems.
Primary Target	Individuals or employees, often tricked into revealing information.	Data and files on infected devices or networks.	Systems, networks, and databases to gain unauthorized access.
Goal	To steal personal or sensitive information like passwords.	To extort money from victims by encrypting their data.	To access, modify, or steal data, or disrupt systems.
User Involvement	High - relies on tricking the user into taking an action.	Medium - often requires user interaction (e.g., clicking a link).	Low - attacker acts directly without needing user interaction.
Example	Fake email claiming to be a bank asking for account information.	Encrypting a computer's data and demanding payment for the key.	Bypassing a company's security to access confidential files.
Preventive Measures	Awareness training, email filtering, and verifying communication.	Regular backups, anti-malware tools, and keeping software updated.	Strong security protocols, firewalls, and regular vulnerability testing.

Types of Cyber Threats:

- **Malware**: Viruses, Trojans, Spyware, Worms.
- **Social Engineering**: Manipulation to gain confidential information.
- **DDoS Attacks**: Disrupting service availability through excessive traffic.
- **Insider Threats**: Employees or associates compromising data security.

Feature	Virus	Trojan	Spyware	Worm
Definition	A type of malware that attaches itself to a host file and spreads when the file is executed.	Malicious software disguised as legitimate, tricking users into downloading or executing.	Software designed to secretly gather user information and send it to a third party.	Standalone malware that replicates itself to spread to other computers or networks.
Primary Objective	To corrupt or destroy data, disrupt operations, or cause harm.	To create a backdoor for unauthorized access or steal data under the guise of normal files.	To monitor user activity, steal sensitive data, or invade privacy.	To spread rapidly across systems and networks, often to cause disruptions or consume resources.
System Harm Level	High	Very High	Moderate to High	High
Impact on System	Can delete, modify, or corrupt files and programs.	Often creates backdoors, steals sensitive data, or installs other malware.	Slows system performance, collects data, and breaches user privacy.	Can crash systems, overwhelm networks, and cause significant performance degradation.

Cybersecurity Policies and Ethics

- **Security Policies**: Guidelines for protecting digital information and systems.
    a. **Acceptable Use Policy (AUP)**: Rules for using IT resources.
    b. **Incident Response Plan**: Steps for managing a security breach.
    c. **Access Control**: Managing who has access to what information.
- **Ethics in Cybersecurity**:
    a. **Respect for Privacy**: Protecting individuals' data.
    b. **Transparency**: Honest and open communication about security practices.
    c. **Accountability**: Holding individuals responsible for their actions.

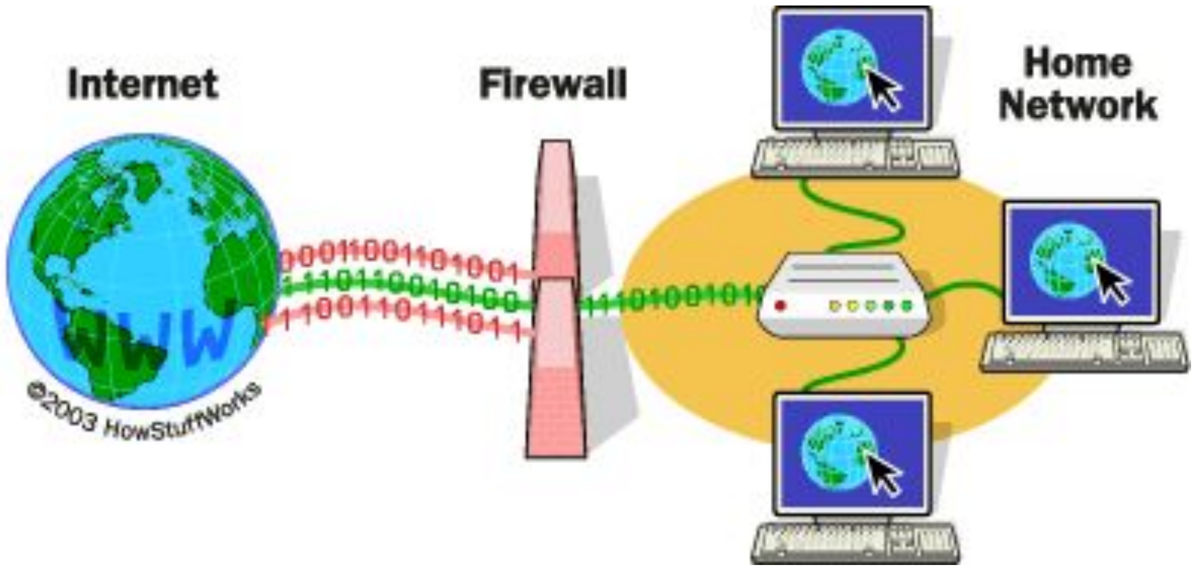
Cybersecurity Tools and Technologies

Popular Tools:

- **Wireshark**: Network protocol analyzer.
- **Metasploit**: Framework for penetration testing.
- **Kali Linux**: Operating system for ethical hacking.
- **Snort**: Intrusion detection system.

Cyber Defense Strategies

- **Best Practices**:
  a. **Firewalls**: Blocking unauthorized access.
  b. **Antivirus Software**: Detecting and removing malware.
  c. **Encryption**: Securing data by converting it into a code.
- **Multi-Factor Authentication (MFA)**: Adding an additional layer of security.



How Firewalls work:

This is an example rule set which defines which connections/packets will be accepted or denied

TABLE I. EXAMPLE FIREWALL RULE SET

Rule no.	Source		Destination		Protocol	Action
	IP	*Port*	*IP*	*Port*		
1	10.10.1.0/24	Any	10.1.0.0/24	Any	TCP	Deny
2	10.10.0.0/16	Any	10.1.0.0/24	Any	TCP	Accept
3	10.10.1.0/24	Any	10.1.0.5/32	Any	TCP	Accept
4	10.10.2.0/24	Any	10.1.0.8/32	Any	TCP	Accept
5	10.10.3.0/24	Any	10.2.0.0/25	80	TCP	Accept
6	10.10.3.0/24	Any	10.2.128.0/25	80	TCP	Accept
7	Any	Any	Any	Any	Any	Deny

How do we define Blockchain?

Formal Definition

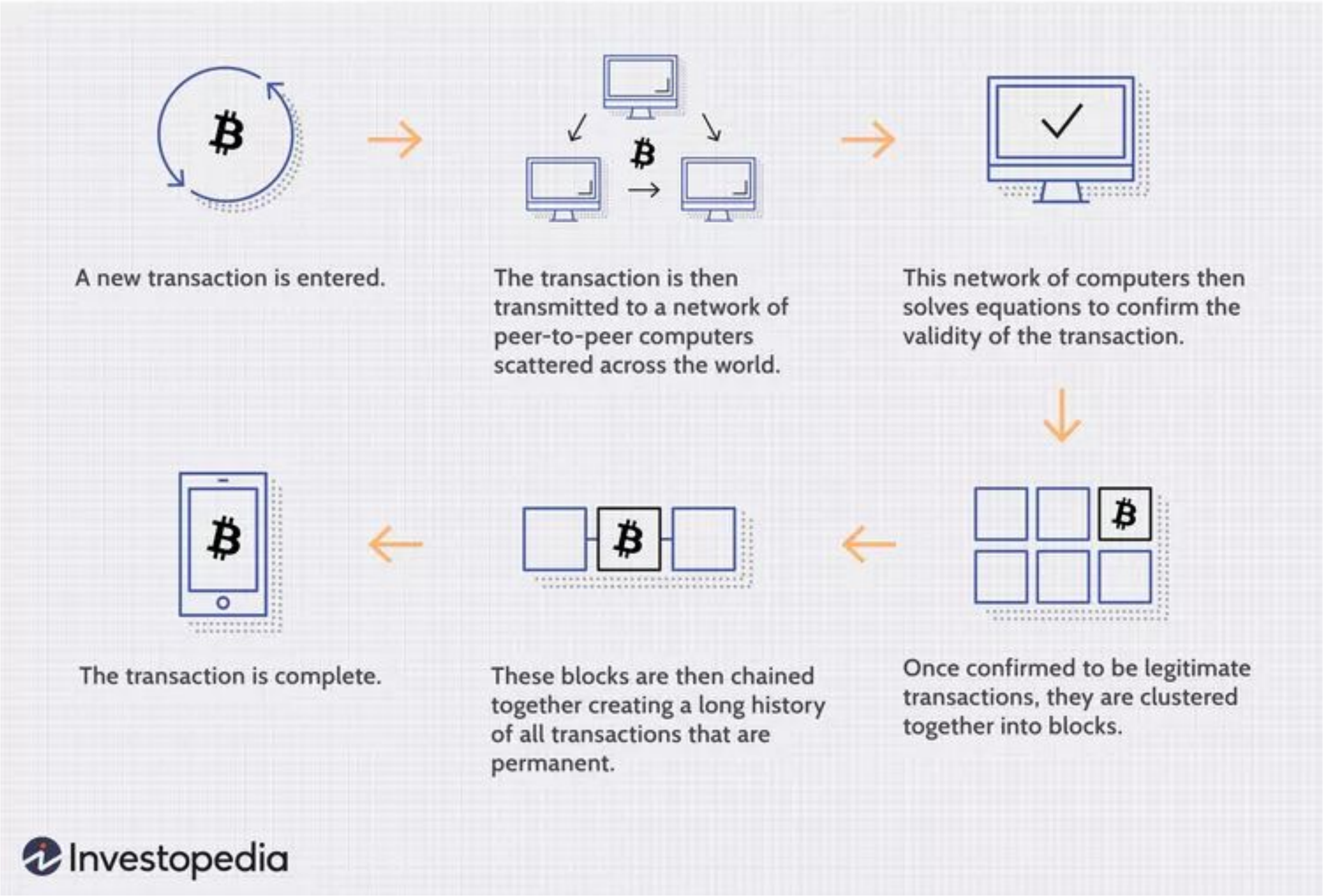
Blockchain is a public ledger consisting of all transactions taken place across a peer-to-peer network.

1. Blockchain is a decentralized, distributed ledger technology that records transactions across multiple computers securely and transparently.

2. Unlike traditional centralized databases, blockchain operates without a central authority making it highly secure and resistant to tampering or fraud.

3. They are best known for their crucial role in cryptocurrency systems, maintaining a secure and decentralized record of transactions, but they are not limited to cryptocurrency uses.

4. Different types of information can be stored on a blockchain, but the most common use has been as a transaction ledger.


A new transaction is entered.
The transaction is then transmitted to a network of peer-to-peer computers scattered across the world.
This network of computers then solves equations to confirm the validity of the transaction.
Once confirmed to be legitimate transactions, they are clustered together into blocks.
These blocks are then chained together creating a long history of all transactions that are permanent.
The transaction is complete.
Investopedia

Key Features of Blockchain

1. **Decentralization** eliminates the need for intermediaries by using peer-to-peer networks.

2. Transactions are visible to all participants so it is **transparent**.

3. Data cannot be altered once recorded. Ensures **Immutability.**

4. Uses cryptographic techniques to **secure** data.

Applications of Blockchain

Biggest application is of Cryptocurrencies, digital currencies like Bitcoin, Ethereum, and stablecoins.

1. Alice wants to send 1 Bitcoin to Bob.
2. Alice creates a transaction using her crypto wallet.
3. The transaction is broadcast to the Bitcoin network.
4. Nodes validate the transaction to ensure it’s legitimate.
5. Miners add the transaction to a block after solving a puzzle (Proof of Work).
6. The block is added to the blockchain, making it permanent.
7. Bob receives the Bitcoin in his wallet after confirmation.

Questions & Answers

That's all folks!

Resources

- **Kevin Knight, Elaine Rich, B. Nair – Artificial Intelligence** – Tata McGraw-Hill Education Pvt. Ltd. (2010)
- **Jiawei Han, Jian Pei, Hanghang Tong – Data Mining Concepts and Techniques** – The Morgan Kaufmann Series in Data Management Systems
- **Russell S , Norvig P – Artificial Intelligence – A Modern Approach,** Second Edition